Supplemental Appendix accompanying *"The Semblance of Success in Nudging Consumers to Pay Down Credit Card Debt"*

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November 2025

Supplemental Appendix Contents:

A. Additional Results for Main Lender

Definitions of Primary Outcomes

1) **Any minimum payment:** Binary outcome for target card. Defined as only paying exactly the minimum (unless that is zero or equal to the full statement balance).
2) **Any full payment:** Binary outcome for target card. Defined as paying the full statement balance (or if no payment is due because there's a zero statement balance).
3) **Any missed payment:** Binary outcome for target card. Defined as paying zero or less than the minimum.
4) **Statement balance net of payments (% statement balance):** Continuous outcome for target card as a measure of credit card debt. Defined as the value of statement balance net of payments as a percent of the value of statement balance. This is the fraction of credit card debt remaining after payments.
5) **Costs (% statement balance):** Continuous outcome for target card. A measure of the costs of borrowing. Defined as the sum of credit card interest and fees as a percentage of statement balance.
6) **Spending (% statement balance):** Continuous outcome for target card. A measure of consumption. Defined as the sum of the value of new credit card transactions that statement cycle as a percentage of statement balance.
7) **Share of credit card portfolio only paying minimum:** Outcome ranging from zero to one. Defined as the proportion of credit cards paying exactly the minimum (unless that is zero or equal to the full balance).

8) **Share of credit card portfolio making full payment:** Outcome ranging from zero to one. Defined as the proportion of credit cards paying the full statement balance (or if no payment is due because there's a zero statement balance).

9) **Share of credit card portfolio missing payment:** Outcome ranging from zero to one. Defined as the proportion of credit cards paying zero or less than the minimum.

10) **Credit card portfolio balances net of payments (% statement balances):** Continuous outcome for credit card portfolio. Defined as the aggregated value of statement balances net of payments across the credit card portfolio as a percent of the aggregated value of statement balances across credit card portfolio. This is the fraction of credit card debt portfolio remaining after payments.

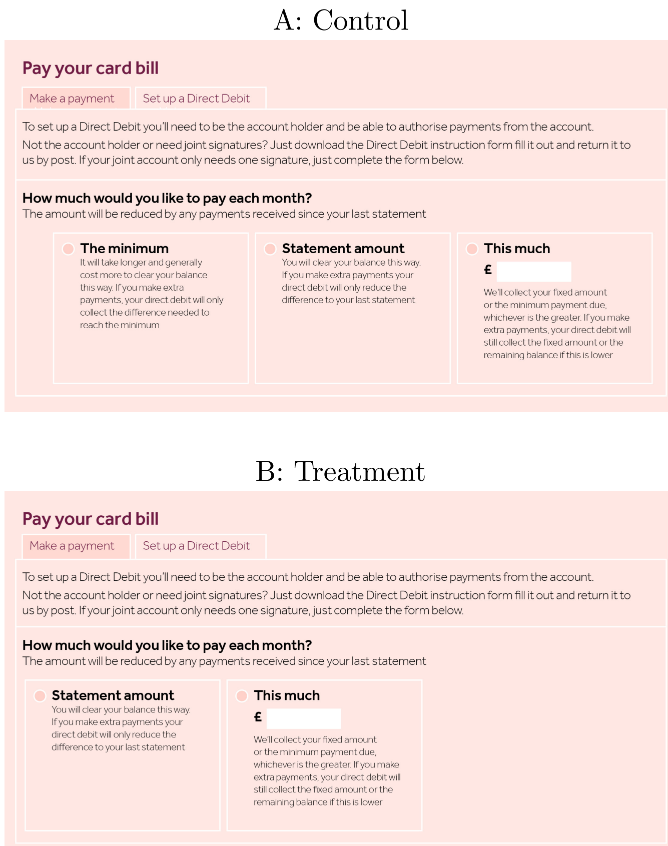

FIGURE A1. AUTOPAY ENROLLMENT CHOICE ARCHITECTURE PRESENTED TO CONTROL (PANEL A) AND TREATMENT (PANEL B) GROUPS

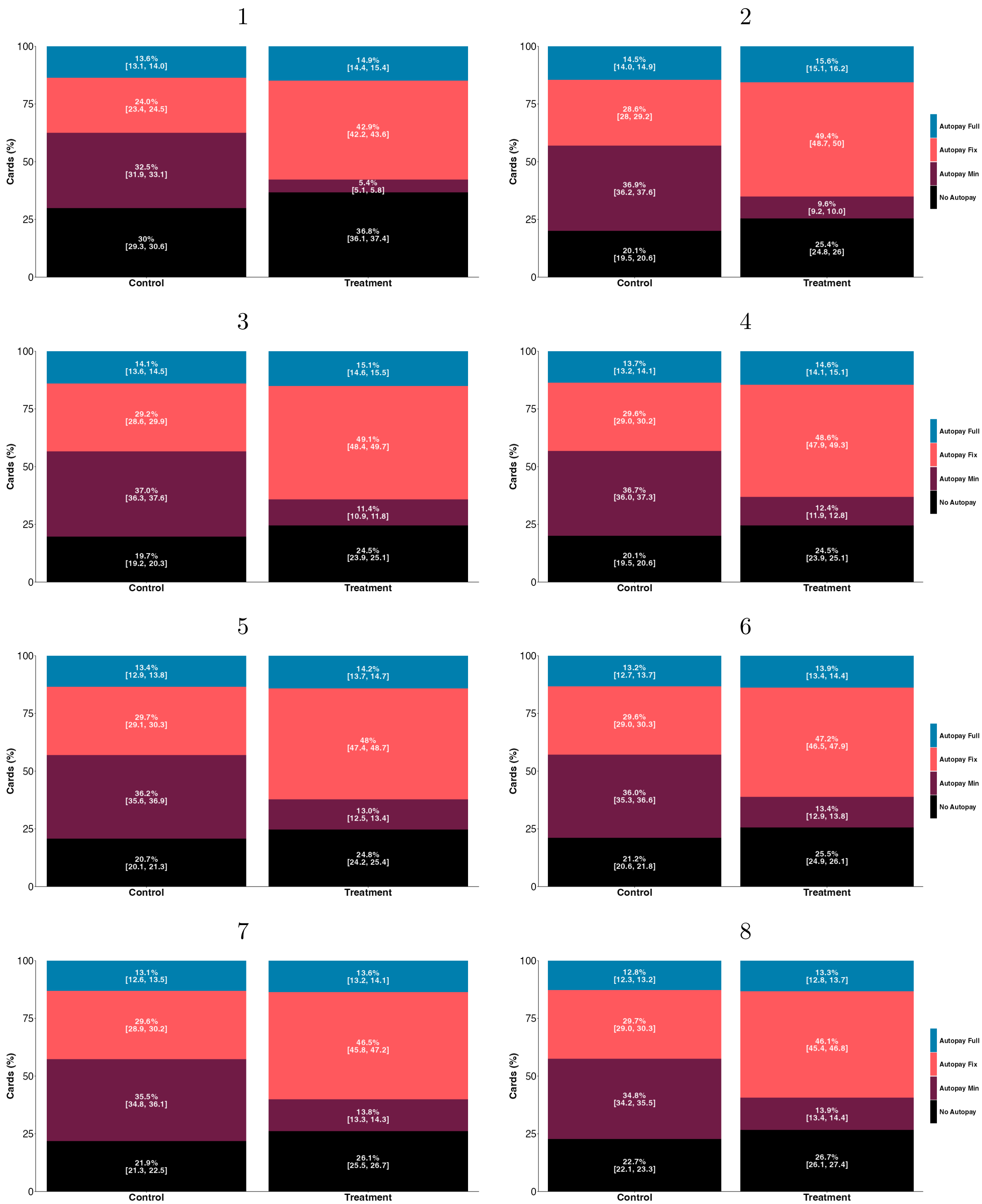

FIGURE A2. AUTOPAY ENROLLMENT FOR CONTROL AND TREATMENT GROUPS, BY STATEMENT CYCLES ONE TO EIGHT

Notes: Numbers display percentage of cards enrolled in each type of Autopay. 95% confidence intervals in [ ]. Cycle 1 is before all treated cards have had 30 days to experience the treatment. Not all cards are observed in cycle 8.

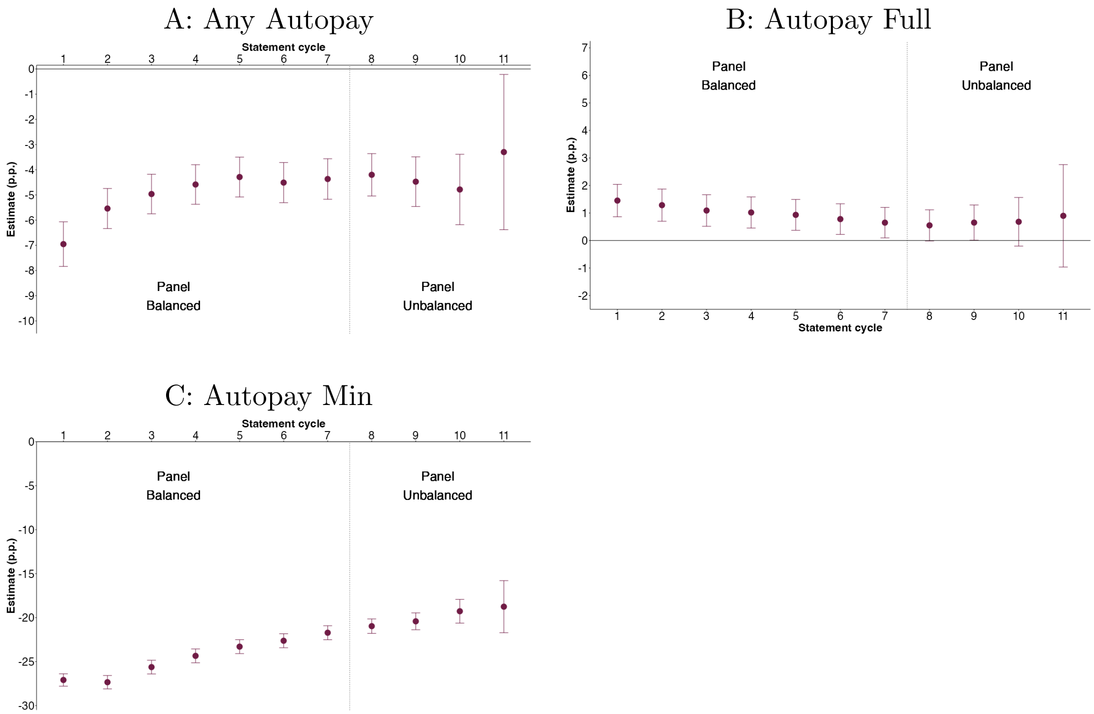

FIGURE A3. AVERAGE TREATMENT EFFECTS ON AUTOPAY ENROLLMENTS, ACROSS 1-11 STATEMENT CYCLES

Notes: Treatment effects from the coefficients (δ_τ) on interaction terms between treatment indicator and statement cycle indicators in the OLS regression specified in Equation 1. The regression outcome in Panel A is any automatic payment enrollment, the outcome in Panel B is any automatic full payment enrollment, and the outcome in Panel C is any automatic minimum payment enrollment. Regressions also include statement cycle fixed effects, year-month fixed effects, and the following controls: Gender, Age, Age squared, Log Estimated Income, Credit Score, Unsecured Debt-to-Income (DTI) Ratio, Any Mortgage Debt, Log Credit Card Credit Limit, Credit Card Purchases Rate, Subprime Credit Card, Any Credit Card Promotional Rate, Any Credit Card Balance Transfer, Credit Card Open Date, Credit Card Statement Day, Any Credit Card Secondary Cardholder. These are all from the time of card origination except for the variables constructed from consumer credit reporting data (Credit Score, DTI Ratio and Any Mortgage Debt), which are from the month preceding card origination. Error bars are 95% confidence intervals. Standard errors are clustered at consumer-level. There are 40,708 credit cards with 368,162 observations.

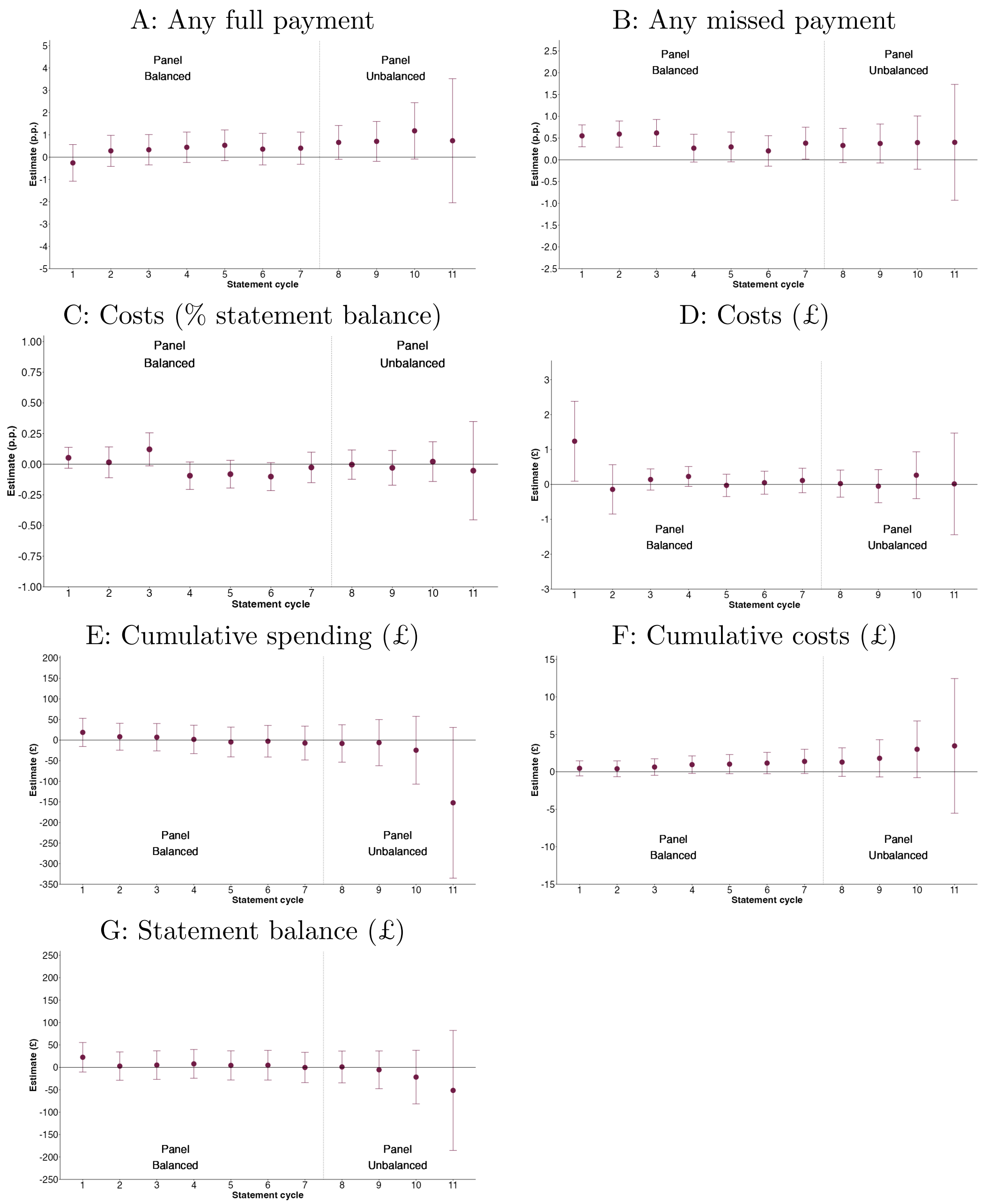

Figure A4. Average treatment effects on outcomes on target card, across 1-11 statement cycles

Notes: Treatment effects from the coefficients (δ_τ) on interaction terms between treatment indicator and statement cycle indicators in the OLS regression specified in Equation 1. The regression outcome in Panel A is pay full balance, outcome in Panel B is pay less than the minimum, the outcome in Panel C is the sum of interest and fees (% statement balance), the outcome in Panel D is the sum of interest and fees (£), the outcome in Panel E is cumulative spending (£), the outcome in Panel E is cumulative costs (£), and the outcome in Panel F is statement balance (£, this is before payments). Regressions also include statement cycle fixed effects, year-month fixed effects, and the following controls: Gender, Age, Age squared, Log Estimated Income, Credit Score, Unsecured Debt-to-Income (DTI) Ratio, Any Mortgage Debt, Log Credit Card Credit Limit, Credit Card Purchases Rate, Subprime Credit Card, Any Credit Card Promotional Rate, Any Credit Card Balance Transfer, Credit Card Open Date, Credit Card Statement Day, Any Credit Card Secondary Cardholder. These are all from the time of card origination except for the variables constructed from consumer credit reporting data (Credit Score, DTI Ratio and Any Mortgage Debt), which are from the month preceding card origination. Error bars are 95% confidence intervals. Standard errors are clustered at consumer-level. There are 40,708 credit cards with 368,162 observations.

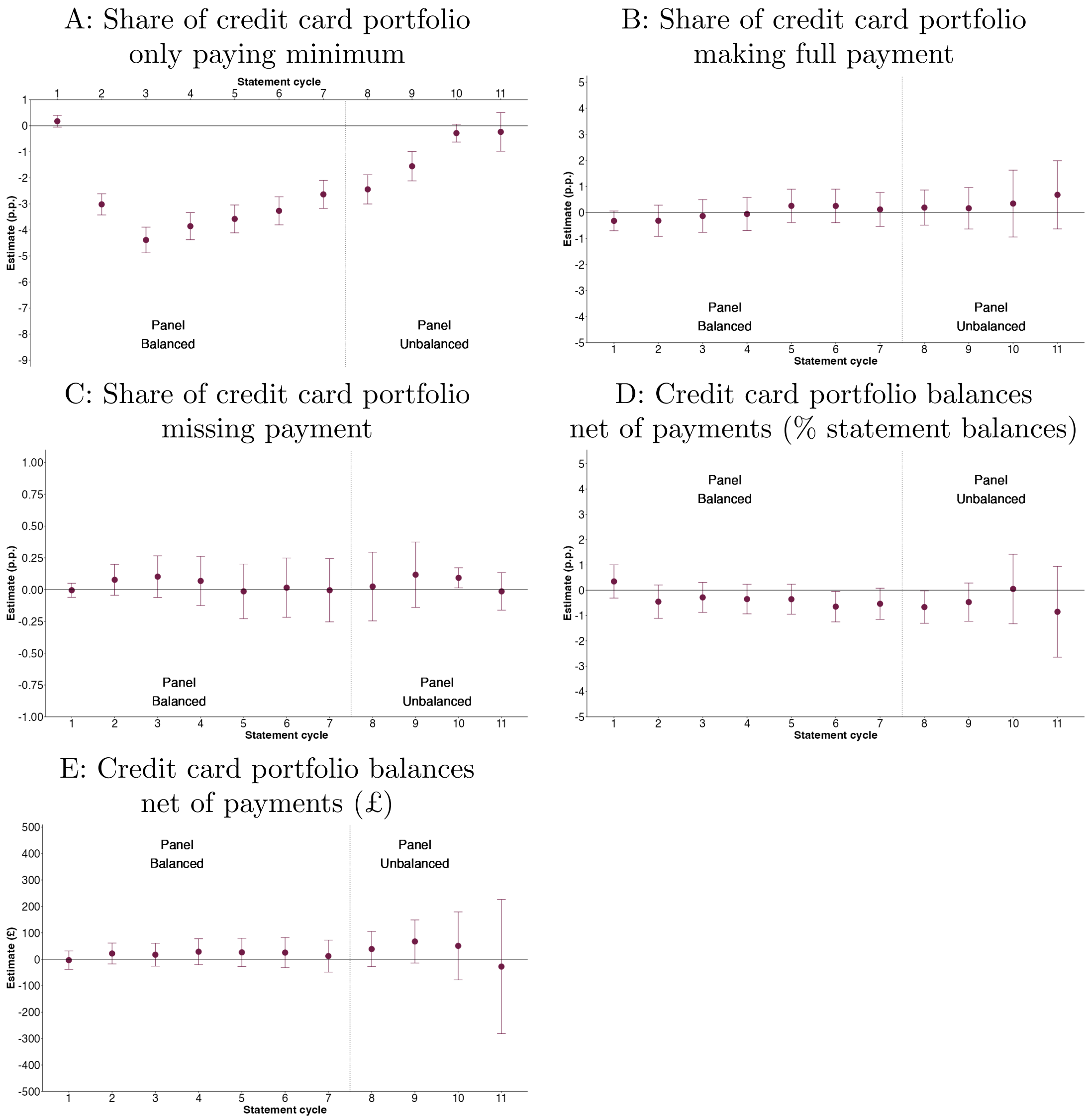

Figure A5. Average treatment effects on credit card portfolio outcomes, across 1-11 statement cycles

Notes: Treatment effects from the coefficients (δ_τ) on interaction terms between treatment indicator and statement cycle indicators in the OLS regression specified in Equation 1. The regression outcome in Panel A is share of credit cards paying the minimum, the outcome in Panel B is share of credit cards paying the full balance, the outcome in Panel C is share of credit cards paying less than the minimum, the outcome in Panel D is credit card portfolio balances net of payments as a percent of statement balances, and the outcome in Panel E is credit card balances net of payments (£). Regressions also include statement cycle fixed effects, year-month fixed effects, and the following controls: Gender, Age, Age squared, Log Estimated Income, Credit Score, Unsecured Debt-to-Income (DTI) Ratio, Any Mortgage Debt, Log Credit Card Credit Limit, Credit Card Purchases Rate, Subprime Credit Card, Any Credit Card Promotional Rate, Any Credit Card Balance Transfer, Credit Card Open Date, Credit Card Statement Day, Any Credit Card Secondary Cardholder. These are all from the time of card origination except for the variables constructed from consumer credit reporting data (Credit Score, DTI Ratio and Any Mortgage Debt), which are from the month preceding card origination. For outcomes constructed from consumer credit reporting data up to eleven dummies for lags of outcomes are included as controls (X_i') for months preceding the start of the experiment. Error bars are 95% confidence intervals. Standard errors are clustered at consumer-level. There are 40,708 credit cards with 368,162 observations.

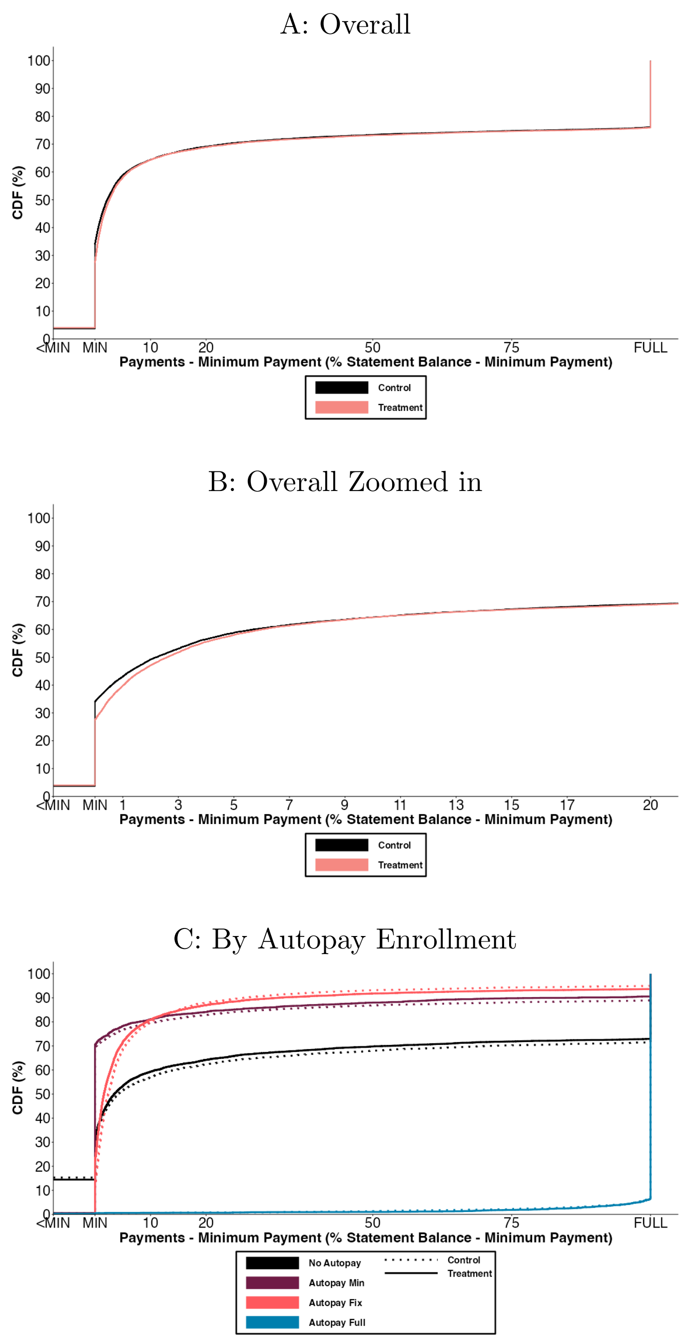

Figure A6. CDF of excess payments (payments − minimum payment, as a % of statement balance − minimum payment) after seven statement cycles

Notes: Figure shows CDFs of payments − minimum payment (% statement balance − minimum payment) measured at the seventh statement cycle. In this outcome, MIN *shows cases where payments equal the minimum payment, cases where payments are less than the minimum are assigned to the* $< MIN$ *group, cases where payments greater than the statement balance are assigned to the* $FULL$ *group. The CDFs show the cumulative fraction of cards within each group. Panel A shows the CDF split by treatment and control groups. Panel B shows the CDF split by treatment and control groups (as in Panel A), but zoomed in to focus on the part of the distribution where the outcome is from* $< MIN$ *to 20%. Panel C shows the CDF split by combinations of treatment status groups and Autopay enrollment groups at the seventh statement cycle. When interpreting Panel C, it is important to note that the sample sizes of cards in each Autopay enrollment status dramatically change as a result of the treatment, as shown in Figure A2. All panels use data on 40,693 credit cards with one observation per card.*

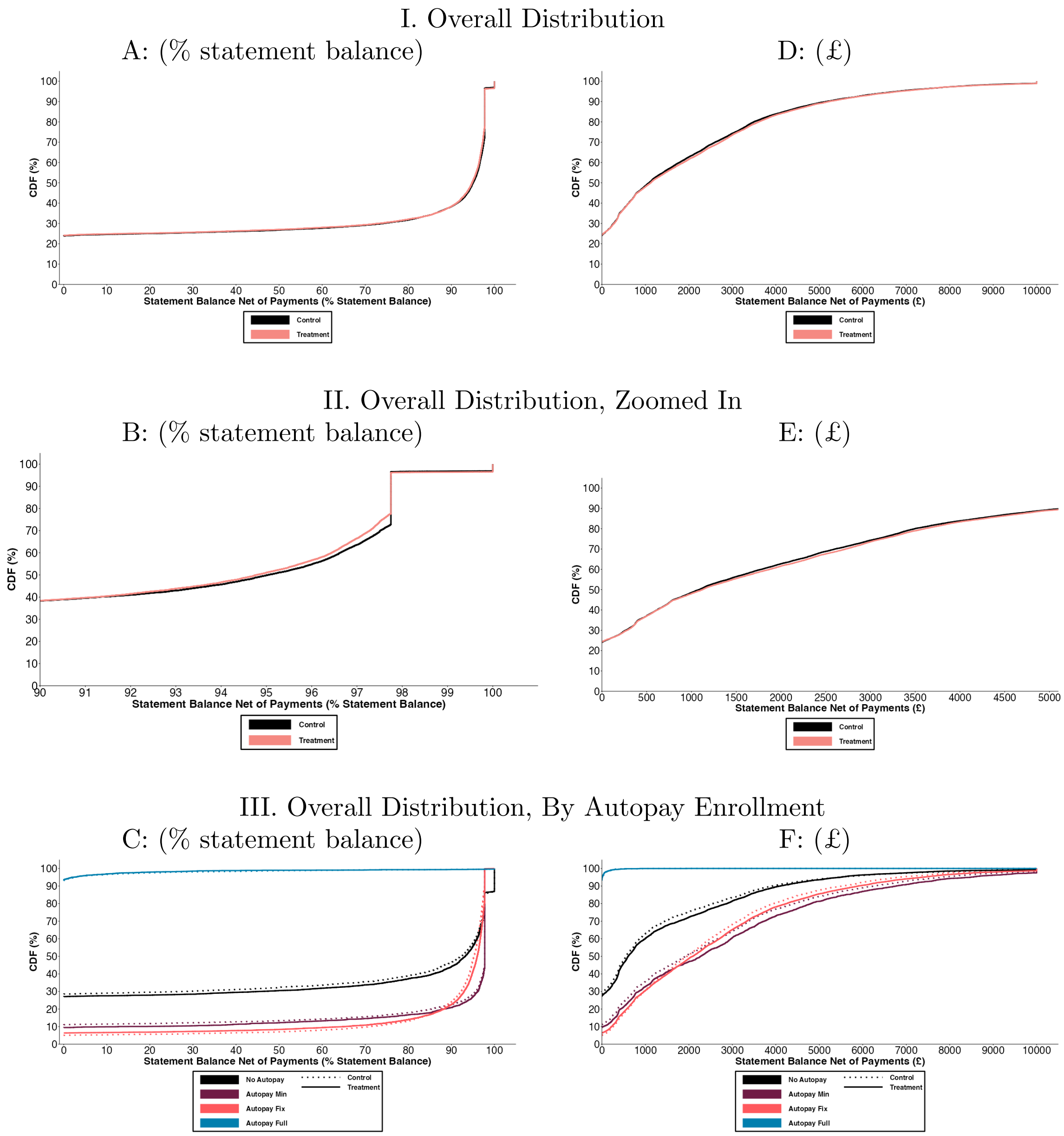

FIGURE A7. CDF OF STATEMENT BALANCE NET OF PAYMENTS AFTER SEVEN STATEMENT CYCLES

Notes: Figure shows CDFs of statement balance net of payments (% statement balance) measured at the seventh statement cycle. The CDFs show the cumulative fraction of cards within each group. Panel A shows the CDF split by treatment and control groups. Panel B shows the CDF split by treatment and control groups (as in Panel A), but zoomed in to focus on the part of the distribution where the outcome is from 90% to 100%. Panel C shows the CDF split by combinations of treatment status groups and Autopay enrollment groups at the seventh statement cycle. Panel D shows CDF split by treatment and control groups. Panel E shows CDF split by treatment and control groups (as in Panel D), but zoomed in to focus on the part of the distribution where the outcome is from £0 to £5,000. Panel F shows the CDF split by combinations of treatment status groups and Autopay enrollment groups at the seventh statement cycle. When interpreting Panels C and F, it is important to note that the sample sizes of cards in each Autopay enrollment status dramatically change as a result of the treatment, as shown in Figure A2. All panels use data on 40,693 credit cards with one observation per card.

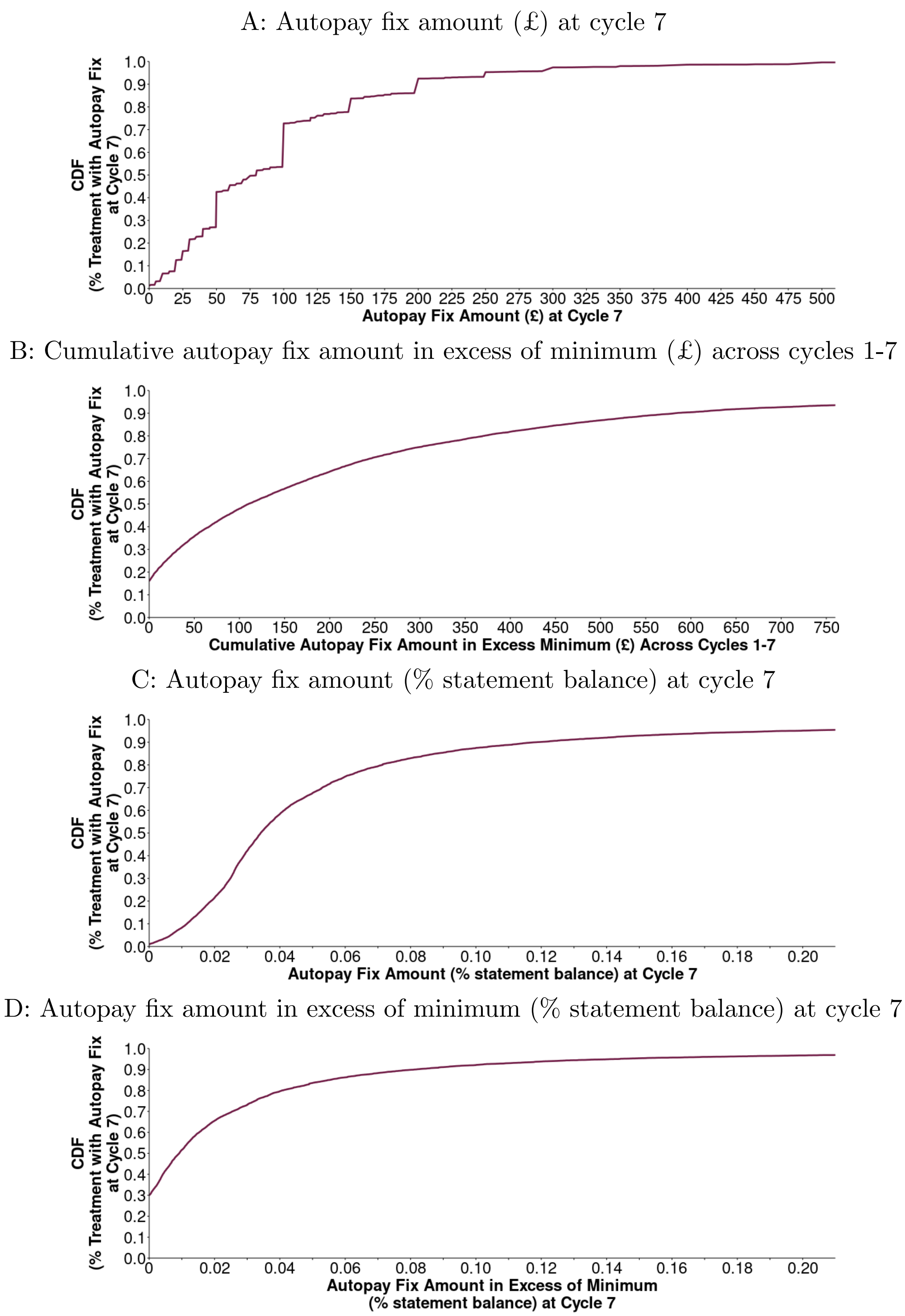

Figure A8. CDF of Autopay Fix payment amounts for those enrolled in Autopay Fix in the treatment group after seven statements

Notes: The x-axes of CDFs are right-censored to ease presentation. The CDFs are calculated for the 9,337 credit cards that are in treatment group and enrolled in Autopay Fix at cycle 7.

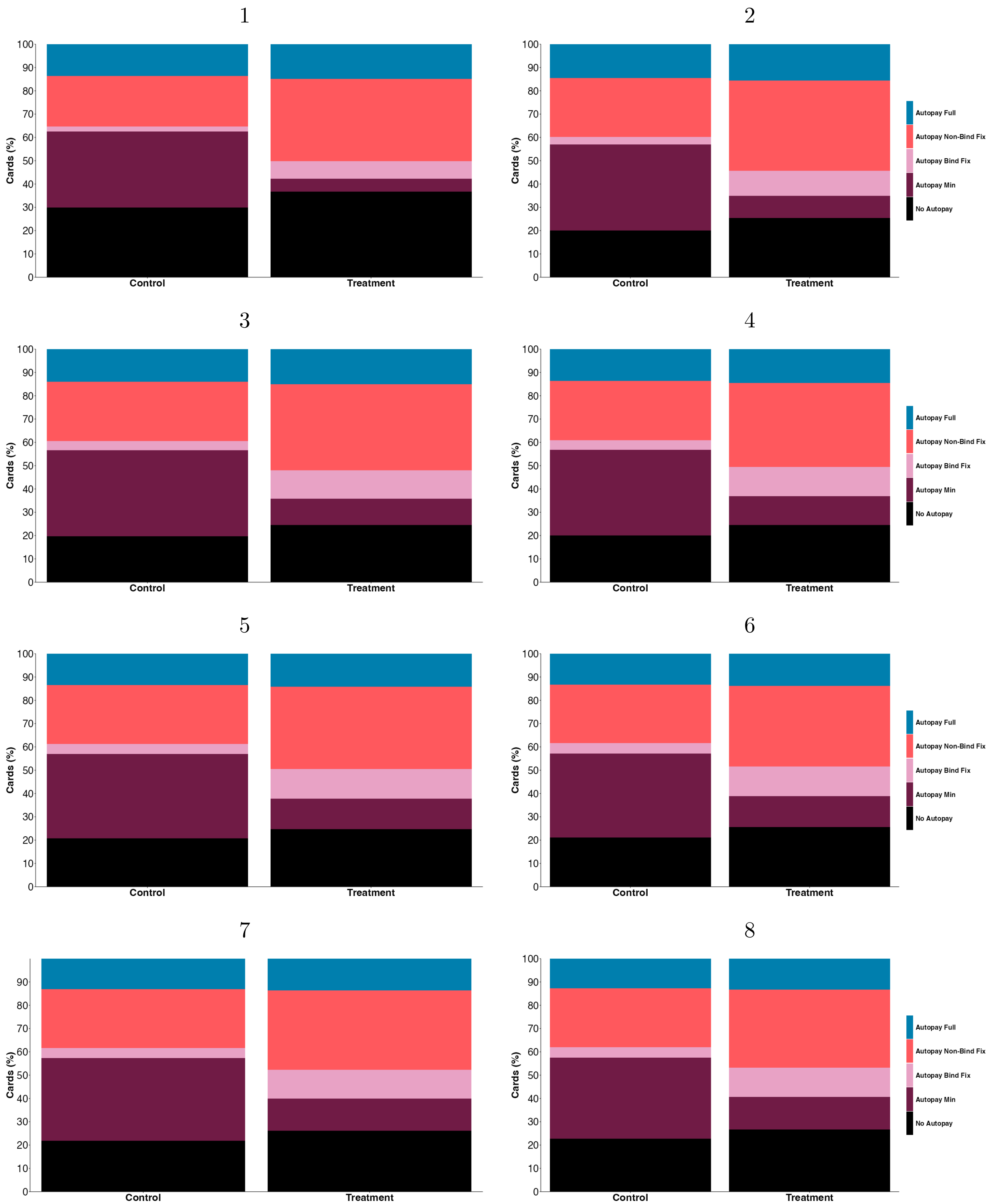

Figure A9. Autopay enrollment - splitting out automatic fixed payments into those that do and do not bind at the minimum payment amount - for control and treatment groups split by statement cycles one to eight

Notes: The numbers display percentage of cards enrolled in each type of Autopay. 95% confidence intervals in [ ].

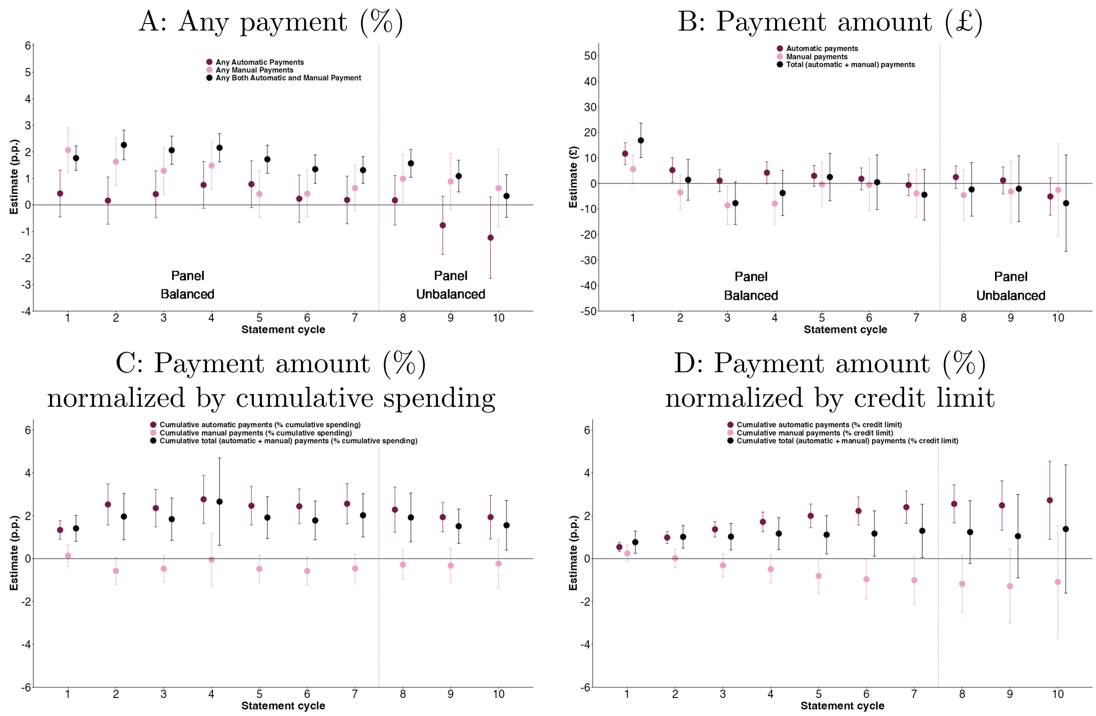

FIGURE A10. AVERAGE TREATMENT EFFECTS ON PAYMENTS, ACROSS 1-10 STATEMENT CYCLES

Notes: Treatment effects from the coefficients (δ_τ) on interaction terms between treatment indicator and statement cycle indicators in the OLS regression specified in Equation 1. Regressions also include statement cycle fixed effects, year-month fixed effects, and the following controls: Gender, Age, Age squared, Log Estimated Income, Credit Score, Unsecured Debt-to-Income (DTI) Ratio, Any Mortgage Debt, Log Credit Card Credit Limit, Credit Card Purchases Rate, Subprime Credit Card, Any Credit Card Promotional Rate, Any Credit Card Balance Transfer, Credit Card Open Date, Credit Card Statement Day, Any Credit Card Secondary Cardholder. These are all from the time of card origination except for the variables constructed from consumer credit reporting data (Credit Score, DTI Ratio and Any Mortgage Debt), which are from the month preceding card origination. Error bars are 95% confidence intervals. Standard errors are clustered at consumer-level. There are 40,708 credit cards with 368,162 observations. Cycle 11 is excluded from figure as, due to few cards being observed in this cycle, the confidence intervals are extremely large such that estimates are uninformative.

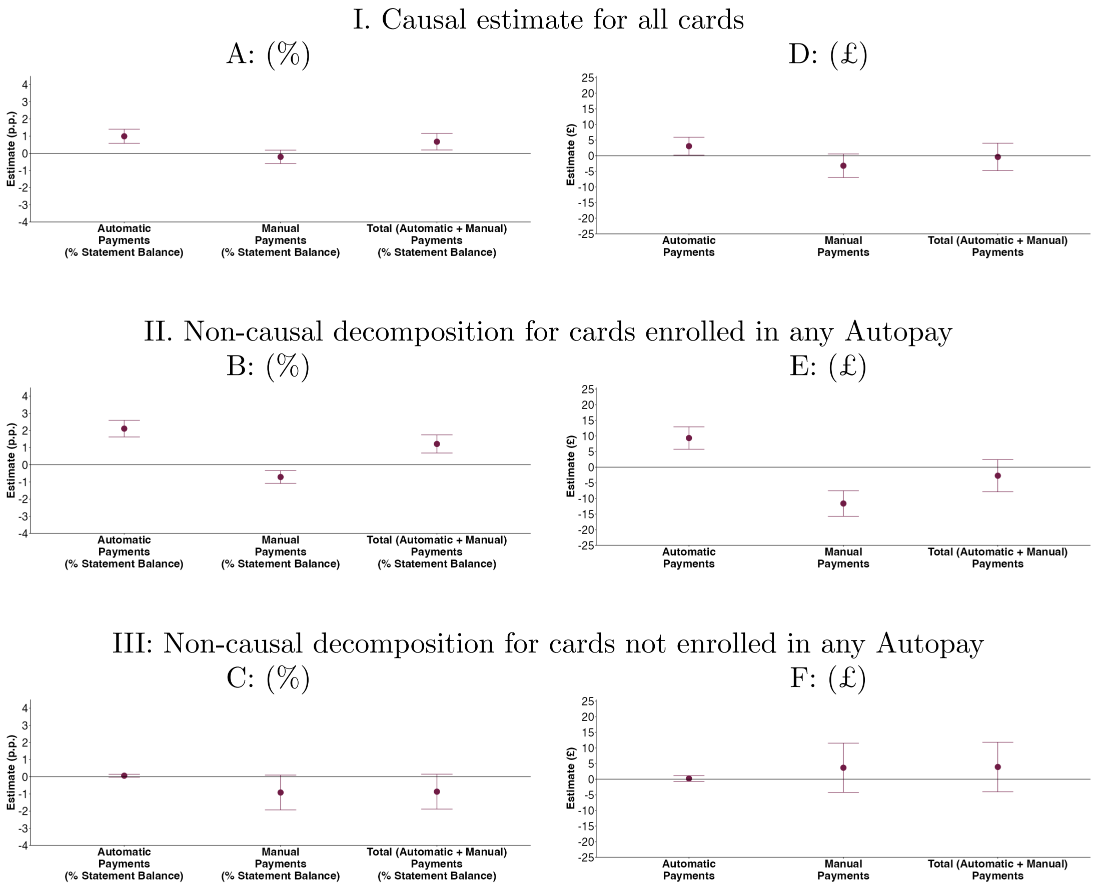

Figure A11. Estimates on payments pooled across all statement cycles, decomposed by any Autopay enrollment after seven statement cycles

Notes: Panels A and D are the causal estimated treatment effects from the coefficient (δ) on the treatment indicator in the OLS regression specified in Equation 2. Each panel (A to F) show the outcomes from three separate regressions where outcomes are: automatic payments, manual payments, and total (automatic +) manual) payments. In Panels A, B, and C these outcomes are all measured as % statement balance. In Panels D, E, and F these outcomes are all measured in £. Regressions also include: statement cycle fixed effects, year-month fixed effects, and the following controls: Gender, Age, Age squared, Log Estimated Income, Credit Score, Unsecured Debt-to-Income (DTI) Ratio, Any Mortgage Debt, Log Credit Card Credit Limit, Credit Card Purchases Rate, Subprime Credit Card, Any Credit Card Promotional Rate, Any Credit Card Balance Transfer, Credit Card Open Date, Credit Card Statement Day, Any Credit Card Secondary Cardholder. These are all from the time of card origination except for the variables constructed from consumer credit reporting data (Credit Score, DTI Ratio and Any Mortgage Debt), which are from the month preceding card origination. Error bars are 95% confidence intervals. Standard errors are clustered at consumer-level. Panels A and D contain 40,708 credit cards with 368,162 observations. Panels B, C, E, and F show the non-causal estimates (δ) from the OLS regression specificed in Equation 3, this has the same specification as explained above except Panels B and E restrict to the subsample of 31,052 credit cards that are enrolled in any Autopay at statement cycle 7, and Panels C and F restrict to the subsample of 9,641 credit cards that are not enrolled in any Autopay at statement cycle 7.

Table A1—Summary statistics

Outcome	Mean	S.D.	P10	P25	P50	P75	P90
Age (years)	36.46	12.44	23	27	34	45	54
Female (% cards)	0.46	0.50	0	0	0	1	1
Credit limit (£)	4356.81	3366.08	660	1,400	3,800	6,300	9,000
Any credit score	0.99	0.12	1	1	1	1	1
Credit score (0-100)	0.65	0.07	0.560	0.610	0.660	0.700	0.740
Purchases rate (%)	22.85	6.11	18.900	18.900	18.900	29.900	34.900
Any estimated income	0.97	0.18	1	1	1	1	1
Estimated income (£)	2437.38	2155.20	899	1,321	1,880	2,816	4,336
Any autopay	0.78	0.41	0	1	1	1	1
Autopay full	0.13	0.34	0	0	0	0	1
Autopay fix	0.30	0.46	0	0	0	1	1
Autopay min	0.35	0.48	0	0	0	1	1
Statement balance (£)	2164.49	2416.30	0	373	1,290	3,274	5,437
Statement balance net of payments (£)	1962.52	2369.65	0	41	1,086	3,070	5,162
Statement balance net of payments (% statement balance)	0.69	0.41	0	0.180	0.950	0.980	0.980
Utilization	0.52	0.37	0	0.200	0.530	0.840	0.980
Any minimum payment	0.30	0.46	0	0	0	1	1
Any full payment	0.24	0.43	0	0	0	0	1
Any missed payment	0.04	0.19	0	0	0	0	0
Cumulative number times paid minimum	2.04	2.63	0	0	0	4	7
Cumulative number times paid in full	1.90	2.56	0	0	1	3	7
Cumulative number times paid less than minimum	0.19	0.76	0	0	0	0	0
6+ times paid minimum	0.19	0.39	0	0	0	0	1
6+ times paid in full	0.18	0.38	0	0	0	0	1
6+ times paid less than minimum	0.01	0.07	0	0	0	0	0
Number of credit cards	2.80	1.90	1	1	2	4	5
Number of credit cards with debt	1.52	1.36	0	1	1	2	3
Credit card portfolio statement balances (£)	3916.96	5142.72	90	626	2,284	5,143	9,734
Credit card portfolio balances net of payments (£)	3431.69	4849.58	0	255	1,851	4,597	8,830

Notes: Summary statistics are calculated for control group (N = 20,609 credit cards) after seventh statement cycle.

Table A2—Minimum Detectable Effect (MDE) sizes at cycle 7 across significance levels 0.005, 0.01, and 0.05 (all assuming 80% power)

Outcome	0.005	0.01	0.05
Any minimum payment	0.0160	0.0150	0.0123
Any full payment	0.0155	0.0145	0.0119
Any missed payment	0.0070	0.0065	0.0053
Statement balance net of payments (% statement balance)	0.0149	0.0140	0.0114
Costs (% statement balance)	0.0023	0.0022	0.0018
Spending (% statement balance)	0.0127	0.0119	0.0098
Share of credit card portfolio only paying minimum	0.0108	0.0101	0.0083
Share of credit card portfolio making full payment	0.0136	0.0127	0.0104
Share of credit card portfolio missing payment	0.0048	0.0045	0.0037
Credit card portfolio balances net of payments (% statement balances)	0.0141	0.0132	0.0108
Any autopay	0.0154	0.0145	0.0119
Autopay full	0.0123	0.0115	0.0095
Autopay fix	0.0176	0.0164	0.0135
Autopay min	0.0156	0.0146	0.0120
Autopay fix exceeding minimum payment amount	0.0165	0.0155	0.0127
Cumulative total payments (£)	63.24	59.23	48.56
Cumulative automatic payments (£)	40.68	38.10	31.24
Cumulative manual payments (£)	52.03	48.73	39.95
Total payments (% statement balance)	0.0130	0.0121	0.0099
Automatic payments (% statement balance)	0.0098	0.0092	0.0075
Manual payments (% statement balance)	0.0103	0.0097	0.0079
Made both automatic and manual payment	0.0094	0.0088	0.0072
Statement balance (£)	87.94	82.36	67.52
Statement balance net of payments (£)	86.26	80.80	66.24
Cumulative spending (£)	106.30	99.56	81.62
Cumulative costs (£)	3.52	3.30	2.71
Spending (£)	17.18	16.09	13.19
Total payments (£)	18.88	17.68	14.50
Automatic payments (£)	8.08	7.57	6.21
Manual payments (£)	17.59	16.47	13.50
Credit card portfolio repayments (£)	40.19	37.64	30.86
Credit card portfolio repayments (% statement balances)	0.01	0.01	0.01
Credit card portfolio statement balances (£)	187.77	175.87	144.18
Credit card portfolio balances net of payments (£)	176.31	165.14	135.38

Table A3—Balance comparison

Outcome	Control	Treatment	Difference (p.p.)	95% C.I.
Age (years)	36.46	36.61	0.14	[-0.10, 0.39]
Female (% cards)	0.4606	0.4612	0.0006	[-0.0091, 0.0103]
Any estimated income	0.9660	0.9630	-0.0030	[-0.0066, 0.0006]
Estimated income (£)	2437.38	2457.50	20.13	[-21.93, 62.19]
Credit limit (£)	4356.81	4429.03	72.22	[6.36, 138.08]
Any credit score	0.9856	0.9834	-0.0023	[-0.0047, 0.0001]
Credit score (0-100)	0.6526	0.6538	0.0012	[-0.0003, 0.0026]
Purchases rate (%)	22.85	22.82	-0.03	[-0.15, 0.09]
Any balance transfer offered	0.2900	0.2976	0.0076	[-0.0013, 0.0164]
Number of credit cards	2.18	2.19	0.02	[-0.02, 0.05]
Number of credit cards with debt	0.90	0.91	0.01	[-0.01, 0.04]
Credit card portfolio statement balances (£)	2364.92	2439.09	74.16	[-0.79, 149.12]
Credit card portfolio balances net of payments (£)	2001.35	2072.53	71.18	[2.59, 139.77]

Notes: N (control) = 20,617 and N (treatment) = 20,091 cards.

Table A4—Average treatment effects for secondary outcomes after seven statement cycles

Outcome	Estimate (s.e.)	95% C.I.	P value	Control mean
Cumulative number times paid in full	0.0192 (0.0201)	[-0.0203, 0.0586]	0.3405	1.9020
Cumulative number times paid minimum	-0.5939 (0.0232)	[-0.6393, -0.5485]	0.0000	2.0444
Cumulative number times paid less than minimum	0.0276 (0.0075)	[0.0129, 0.0424]	0.0002	0.1892
Cumulative total payments (£)	6.68 (16.19)	[-25.06, 38.41]	0.6800	1277.27
Cumulative automatic payments (£)	27.30 (10.35)	[7.01, 47.59]	0.0084	573.79
Cumulative manual payments (£)	-18.87 (13.97)	[-46.25, 8.50]	0.1766	711.97
Total payments (% statement balance)	0.0060 (0.0032)	[-0.0002, 0.0123]	0.0579	0.2271
Automatic payments (% statement balance)	0.0072 (0.0025)	[0.0023, 0.0122]	0.0040	0.1101
Manual payments (% statement balance)	-0.0005 (0.0028)	[-0.0061, 0.0050]	0.8477	0.1212
Made both automatic and manual payment	0.0131 (0.0026)	[0.0080, 0.0182]	0.0000	0.0672
Statement balance (£)	-0.33 (17.24)	[-34.11, 33.46]	0.9848	2164.49
Statement balance net of payments (£)	4.11 (17.22)	[-29.64, 37.85]	0.8115	1962.52
Utilization	0.0002 (0.0032)	[-0.0061, 0.0064]	0.9604	0.5223
Cumulative spending (£)	-7.23 (20.95)	[-48.29, 33.83]	0.7300	3186.19
Cumulative costs (£)	1.39 (0.83)	[-0.23, 3.01]	0.0924	76.02
Spending (£)	-9.84 (4.99)	[-19.61, -0.07]	0.0485	193.24
Total payments (£)	-4.44 (5.07)	[-14.38, 5.51]	0.3820	201.98
Automatic payments (£)	-0.6123 (2.1195)	[-4.7665, 3.5419]	0.7727	86.9490
Manual payments (£)	-3.90 (4.79)	[-13.29, 5.48]	0.4152	116.38
Credit card portfolio repayments (£)	9.11 (9.39)	[-9.29, 27.51]	0.3318	485.70
Credit card portfolio repayments (% statement balances)	0.00 (0.00)	[-0.00, 0.01]	0.5730	0.26
Credit card portfolio statement balances (£)	23.65 (31.15)	[-37.42, 84.71]	0.4479	3916.96
Credit card portfolio balances net of payments (£)	12.06 (30.92)	[-48.55, 72.66]	0.6966	3431.69

Notes: Table shows average treatment effects after seven statement cycles. Each row of table shows estimates from separate regressions with different outcomes. Estimated treatment effects from the coefficient (δ_7) on interaction terms between treatment indicator and the seventh statement cycle indicator in the OLS regression specified in Equation 1. Regressions also include: interactions between treatment indicator and other statement cycles, statement cycle fixed effects, year-month fixed effects, and the following controls: Gender, Age, Age squared, Log Estimated Income, Credit Score, Unsecured Debt-to-Income (DTI) Ratio, Any Mortgage Debt, Log Credit Card Credit Limit, Credit Card Purchases Rate, Subprime Credit Card, Any Credit Card Promotional Rate, Any Credit Card Balance Transfer, Credit Card Open Date, Credit Card Statement Day, Any Credit Card Secondary Cardholder. These are all from the time of card origination except for the variables constructed from consumer credit reporting data (Credit Score, DTI Ratio and Any Mortgage Debt), which are from the month preceding card origination. For outcomes constructed from consumer credit reporting data up to eleven dummies for lags of outcomes are included as controls (X_i') for months preceding the start of the experiment. Standard errors are clustered at consumer-level. There are 40,708 credit cards with 368,162 observations.

Table A5—Average treatment effects for outcomes pooled across all statement cycles

Outcome	Estimate (s.e.)	95% C.I.	P value	Control mean
Any minimum payment	-0.0807 (0.0033)	[-0.0871, -0.0742]	0.0000	0.2943
Any full payment	0.0041 (0.0028)	[-0.0015, 0.0096]	0.1489	0.2658
Any missed payment	0.0040 (0.0011)	[0.0019, 0.0062]	0.0002	0.0297
Statement balance net of payments (% statement balance)	-0.0056 (0.0027)	[-0.0109, -0.0003]	0.0380	0.6692
Costs (% statement balance)	-0.0001 (0.0002)	[-0.0006, 0.0003]	0.5166	0.0109
Spending (% statement balance)	0.0012 (0.0020)	[-0.0027, 0.0052]	0.5430	0.2918
Share of credit card portfolio only paying minimum	-0.0266 (0.0017)	[-0.0298, -0.0233]	0.0000	0.1631
Share of credit card portfolio making full payment	0.0002 (0.0023)	[-0.0043, 0.0048]	0.9190	0.5150
Share of credit card portfolio missing payment	0.0004 (0.0007)	[-0.0009, 0.0017]	0.5400	0.0144
Credit card portfolio balances net of payments (% statement balances)	-0.0036 (0.0022)	[-0.0079, 0.0006]	0.0967	0.6245
Statement balance (£)	3.59 14.94	[-25.70, 32.87]	0.8103	2049.8420
Statement balance net of payments (£)	3.98 14.92	[-25.26, 33.21]	0.7897	1862.3909
Spending (£)	-2.46 2.65	[-7.66, 2.74]	0.3544	395.5314
Costs (£)	0.19 0.12	[-0.04, 0.42]	0.1146	10.6055
Total payments (£)	-0.39 2.24	[-4.78, 4.00]	0.8611	187.4512
Automatic payments (£)	3.0544 (1.4627)	[0.1874, 5.9214]	0.0368	82.6856
Manual payments (£)	-3.21 1.93	[-6.99, 0.57]	0.0962	105.9518
Any Manual Payment	0.0107 (0.0034)	[0.0041, 0.0173]	0.0015	0.3146
Total payments (% statement balance)	0.0067 (0.0025)	[0.0019, 0.0115]	0.0061	0.2346
Automatic payments (% statement balance)	0.0099 (0.0021)	[0.0058, 0.0140]	0.0000	0.1121
Manual payments (% statement balance)	-0.0021 (0.0020)	[-0.0060, 0.0018]	0.2922	0.1268
Credit card portfolio statement balances (£)	30.60 22.28	[-13.06, 74.26]	0.1696	3506.8973
Credit card portfolio balances net of payments (£)	24.99 22.03	[-18.19, 68.17]	0.2567	2961.2714
Credit card portfolio repayments (£)	4.07 4.33	[-4.42, 12.55]	0.3474	545.7112

Notes: Table shows average treatment effects pooled across statement cycles. Each row of table shows estimates from separate regressions with different outcomes. Estimated treatment effects from the coefficient (δ) on the treatment indicator in the OLS regression specified in Equation 2. Regressions also include: statement cycle fixed effects, year-month fixed effects, and the following controls: Gender, Age, Age squared, Log Estimated Income, Credit Score, Unsecured Debt-to-Income (DTI) Ratio, Any Mortgage Debt, Log Credit Card Credit Limit, Credit Card Purchases Rate, Subprime Credit Card, Any Credit Card Promotional Rate, Any Credit Card Balance Transfer, Credit Card Open Date, Credit Card Statement Day, Any Credit Card Secondary Cardholder. These are all from the time of card origination except for the variables constructed from consumer credit reporting data (Credit Score, DTI Ratio and Any Mortgage Debt), which are from the month preceding card origination. Standard errors are clustered at consumer-level. There are 40,708 credit cards with 368,162 observations.

Table A6—Average treatment effects for tertiary arrears outcomes pooled across all statement cycles

Outcome	Estimate (s.e.)	95% C.I.	P value	Control mean
Any missed payment	0.0040 (0.0011)	[0.0019, 0.0062]	0.0002	0.0297
Arrears 1+ payments behind	0.0031 (0.0010)	[0.0011, 0.0051]	0.0024	0.0267
Arrears 2+ payments behind	0.0004 (0.0007)	[-0.0009, 0.0018]	0.5476	0.0110
Arrears 3+ payments behind	0.0002 (0.0005)	[-0.0009, 0.0012]	0.7677	0.0071
Share of credit card portfolio missing payment	0.0004 (0.0007)	[-0.0009, 0.0017]	0.5400	0.0144

Notes: Table shows average treatment effects pooled across statement cycles. Each row of table shows estimates from separate regressions with different outcomes. The first row is our 3rd primary outcome: defined as paying zero or less than the minimum due (on the "target" card in the experiment). The last row is our ninth primary outcome: defined as the proportion of credit cards paying zero or less than the minimum due (constructed from consumer credit reporting data containing the portfolio of credit card held). All other rows show effects for non-primary outcomes for the card in the experiment: standard industry point-in-time measures for the number of payments in arrears was when payments became due. Estimated treatment effects from the coefficient (δ) on the treatment indicator in the OLS regression specified in Equation 2. Regressions also include: statement cycle fixed effects, year-month fixed effects, and the following controls: Gender, Age, Age squared, Log Estimated Income, Credit Score, Unsecured Debt-to-Income (DTI) Ratio, Any Mortgage Debt, Log Credit Card Credit Limit, Credit Card Purchases Rate, Subprime Credit Card, Any Credit Card Promotional Rate, Any Credit Card Balance Transfer, Credit Card Open Date, Credit Card Statement Day, Any Credit Card Secondary Cardholder. These are all from the time of card origination except for the variables constructed from consumer credit reporting data (Credit Score, DTI Ratio and Any Mortgage Debt), which are from the month preceding card origination. For outcomes constructed from consumer credit reporting data up to eleven dummies for lags of outcomes are included as controls (X_i') for months preceding the start of the experiment. Standard errors are clustered at consumer-level. There are 40,708 credit cards with 368,162 observations.

Table A7—Describing compliers (no autopay)

Outcome	Group	Share	Mean	S.E.
Credit score (0-100)	All	1	0.6431	0.0006
Credit score (0-100)	Always Takers (No Autopay)	0.2189	0.6249	0.0015
Credit score (0-100)	Compliers (Autopay to No Autopay)	0.0418	0.6028	0.0135
Credit score (0-100)	Never Takers (Autopay)	0.7393	0.6507	0.0009
Estimated income (£)	All	1	2434.08	10.79
Estimated income (£)	Always Takers (No Autopay)	0.2189	2035.87	26.97
Estimated income (£)	Compliers (Autopay to No Autopay)	0.0418	2038.12	260.09
Estimated income (£)	Never Takers (Autopay)	0.7393	2574.37	18.55
Unsecured Debt to Income (DTI) Ratio	All	1	3.3030	0.0262
Unsecured Debt to Income (DTI) Ratio	Always Takers (No Autopay)	0.2189	3.0764	0.0721
Unsecured Debt to Income (DTI) Ratio	Compliers (Autopay to No Autopay)	0.0418	2.7169	0.6232
Unsecured Debt to Income (DTI) Ratio	Never Takers (Autopay)	0.7393	3.4032	0.0452
Age (years)	All	1	36.53	0.06
Age (years)	Always Takers (No Autopay)	0.2189	33.38	0.18
Age (years)	Compliers (Autopay to No Autopay)	0.0418	31.53	1.55
Age (years)	Never Takers (Autopay)	0.7393	37.75	0.10
Female (% cards)	All	1	0.4609	0.0026
Female (% cards)	Always Takers (No Autopay)	0.2189	0.4683	0.0074
Female (% cards)	Compliers (Autopay to No Autopay)	0.0418	0.5005	0.0606
Female (% cards)	Never Takers (Autopay)	0.7393	0.4564	0.0042
Credit limit (£)	All	1	4347.77	16.67
Credit limit (£)	Always Takers (No Autopay)	0.2189	3260.20	44.10
Credit limit (£)	Compliers (Autopay to No Autopay)	0.0418	2572.29	434.34
Credit limit (£)	Never Takers (Autopay)	0.7393	4770.16	29.14
Purchases rate (%)	All	1	22.83	0.03
Purchases rate (%)	Always Takers (No Autopay)	0.2189	24.82	0.10
Purchases rate (%)	Compliers (Autopay to No Autopay)	0.0418	24.41	0.73
Purchases rate (%)	Never Takers (Autopay)	0.7393	22.15	0.04
Number of credit cards	All	1	2.18	0.01
Number of credit cards	Always Takers (No Autopay)	0.2189	1.71	0.02
Number of credit cards	Compliers (Autopay to No Autopay)	0.0418	1.62	0.23
Number of credit cards	Never Takers (Autopay)	0.7393	2.36	0.02
Number of credit cards with debt	All	1	0.91	0.01
Number of credit cards with debt	Always Takers (No Autopay)	0.2189	0.66	0.01
Number of credit cards with debt	Compliers (Autopay to No Autopay)	0.0418	0.65	0.13
Number of credit cards with debt	Never Takers (Autopay)	0.7393	0.99	0.01
Credit card portfolio balances net of payments (£)	All	1	2036.48	17.70
Credit card portfolio balances net of payments (£)	Always Takers (No Autopay)	0.2189	1290.67	39.18
Credit card portfolio balances net of payments (£)	Compliers (Autopay to No Autopay)	0.0418	826.69	436.51
Credit card portfolio balances net of payments (£)	Never Takers (Autopay)	0.7393	2325.70	30.56
Non-Mortgage Debt Value (£)	All	1	7132.58	51.22
Non-Mortgage Debt Value (£)	Always Takers (No Autopay)	0.2189	5726.35	128.15
Non-Mortgage Debt Value (£)	Compliers (Autopay to No Autopay)	0.0418	6668.40	1198.62
Non-Mortgage Debt Value (£)	Never Takers (Autopay)	0.7393	7575.25	84.05

Notes: Table describes consumers by their observable characteristics at the time of card opening (or before for variables constructed from consumer credit reporting data). We use the approach of Marbach and Hangartner (2020); Hangartner et al. (2021) to estimate the characteristics of three consumer types (as summarized in table below): "Always Takers" who do not receive the treatment and do enroll in any Autopay; "Compliers" who would have enrolled in any Autopay without the treatment but with the treatment do not enroll in any Autopay; and "Never Takers" who receive the treatment and do enroll in any Autopay. These are under the assumption of monotonicity such that the treatment does not make consumers go from no Autopay enrollment to any Autopay enrollment (i.e., No "Defiers"). Autopay enrollment is measured at the seventh statement cycle. N is 40,708 consumers.

	CONTROL (0)	**TREATMENT (1)**
AUTOPAY (0)	Compliers & Never Takers	Never Takers ~~& Defiers~~
NO AUTOPAY (1)	Always Takers ~~& Defiers~~	Compliers & Always Takers

Table A8—Describing compliers (no minimum payment)

Outcome	Group	Share	Mean	S.E.
Credit score (0-100)	All	1	0.6431	0.0005
Credit score (0-100)	Always Takers (No Min Pay)	0.6988	0.6476	0.0009
Credit score (0-100)	Compliers (Min Pay to No Min Pay)	0.0689	0.6412	0.0079
Credit score (0-100)	Never Takers (Min Pay)	0.2323	0.6301	0.0017
Estimated income (£)	All	1	2434.08	10.92
Estimated income (£)	Always Takers (No Min Pay)	0.6988	2386.67	17.44
Estimated income (£)	Compliers (Min Pay to No Min Pay)	0.0689	2619.91	150.66
Estimated income (£)	Never Takers (Min Pay)	0.2323	2521.56	31.71
Unsecured Debt to Income (DTI) Ratio	All	1	3.3030	0.0259
Unsecured Debt to Income (DTI) Ratio	Always Takers (No Min Pay)	0.6988	2.8477	0.0420
Unsecured Debt to Income (DTI) Ratio	Compliers (Min Pay to No Min Pay)	0.0689	3.9706	0.3876
Unsecured Debt to Income (DTI) Ratio	Never Takers (Min Pay)	0.2323	4.4747	0.0930
Age (years)	All	1	36.53	0.06
Age (years)	Always Takers (No Min Pay)	0.6988	36.37	0.11
Age (years)	Compliers (Min Pay to No Min Pay)	0.0689	38.73	0.90
Age (years)	Never Takers (Min Pay)	0.2323	36.37	0.16
Female (% cards)	All	1	0.4609	0.0024
Female (% cards)	Always Takers (No Min Pay)	0.6988	0.4607	0.0042
Female (% cards)	Compliers (Min Pay to No Min Pay)	0.0689	0.4704	0.0359
Female (% cards)	Never Takers (Min Pay)	0.2323	0.4585	0.0074
Credit limit (£)	All	1	4347.77	16.82
Credit limit (£)	Always Takers (No Min Pay)	0.6988	3995.30	26.66
Credit limit (£)	Compliers (Min Pay to No Min Pay)	0.0689	4421.98	237.10
Credit limit (£)	Never Takers (Min Pay)	0.2323	5386.20	54.22
Purchases rate (%)	All	1	22.83	0.03
Purchases rate (%)	Always Takers (No Min Pay)	0.6988	23.16	0.05
Purchases rate (%)	Compliers (Min Pay to No Min Pay)	0.0689	23.22	0.44
Purchases rate (%)	Never Takers (Min Pay)	0.2323	21.73	0.07
Number of credit cards	All	1	2.18	0.01
Number of credit cards	Always Takers (No Min Pay)	0.6988	1.99	0.01
Number of credit cards	Compliers (Min Pay to No Min Pay)	0.0689	2.43	0.13
Number of credit cards	Never Takers (Min Pay)	0.2323	2.69	0.03
Number of credit cards with debt	All	1	0.91	0.01
Number of credit cards with debt	Always Takers (No Min Pay)	0.6988	0.74	0.01
Number of credit cards with debt	Compliers (Min Pay to No Min Pay)	0.0689	1.25	0.08
Number of credit cards with debt	Never Takers (Min Pay)	0.2323	1.30	0.02
Credit card portfolio balances net of payments (£)	All	1	2036.48	17.62
Credit card portfolio balances net of payments (£)	Always Takers (No Min Pay)	0.6988	1494.74	23.53
Credit card portfolio balances net of payments (£)	Compliers (Min Pay to No Min Pay)	0.0689	2875.48	251.62
Credit card portfolio balances net of payments (£)	Never Takers (Min Pay)	0.2323	3417.46	65.75
Non-Mortgage Debt Value (£)	All	1	7132.58	49.33
Non-Mortgage Debt Value (£)	Always Takers (No Min Pay)	0.6988	5972.58	74.66
Non-Mortgage Debt Value (£)	Compliers (Min Pay to No Min Pay)	0.0689	9101.07	674.35
Non-Mortgage Debt Value (£)	Never Takers (Min Pay)	0.2323	10038.62	170.11

Notes: Table describes consumers by their observable characteristics at the time of card opening (or before for variables constructed from consumer credit reporting data). We use the approach of Marbach and Hangartner (2020); Hangartner et al. (2021) to estimate the characteristics of three consumer types: "Always Takers" who do not receive the treatment and only pay exactly the minimum payment; "Compliers" who would have only paid exactly the minimum payment without the treatment but with the treatment do not pay exactly the minimum payment (i.e., they may pay more or less); and "Never Takers" who receive the treatment and only pay exactly the minimum payment. These are under the assumption of monotonicity such that the treatment does not make consumers go from not paying exactly the minimum to paying exactly the minimum payment (i.e., No "Defiers"). Minimum payment is measured at the seventh statement cycle. N is 40,708 consumers.

	CONTROL (0)	**TREATMENT (1)**
MINIMUM PAYMENT (0)	Compliers & Never Takers	Never Takers ~~& Defiers~~
NO MINIMUM PAYMENT (1)	Always Takers ~~& Defiers~~	Compliers & Always Takers

Table A9—Lee bounds for average treatment effects for outcomes after seven statement cycles

Outcome	Lower Bound	Upper Bound
Any minimum payment	-0.1057	-0.0532
Any full payment	-0.0028	0.0360
Any missed payment	-0.0009	0.0036
Statement balance net of payments (% statement balance)	-0.0338	0.0059
Costs (% statement balance)	-0.0012	0.0041
Spending (% statement balance)	-0.0020	0.0414
Share of credit card portfolio only paying minimum	-0.0448	0.0060
Share of credit card portfolio making full payment	-0.0188	0.0239
Share of credit card portfolio missing payment	-0.0013	0.0054
Credit card portfolio balances net of payments (% statement balances)	-0.0309	0.0021
Statement balance (£)	-73.19	254.99
Statement balance net of payments (£)	-51.60	262.37
Costs (£)	-0.30	2.99
Cumulative costs (£)	-1.32	13.49
Spending (£)	-15.71	85.84
Cumulative spending (£)	-97.60	310.84
Arrears 1+ payments behind	-0.0002	0.0090
Arrears 2+ payments behind	0.0004	0.0009
Arrears 3+ payments behind	0.0003	0.0005
Any Autopay Payment	0.0080	0.0585
Any Manual Payment	-0.0323	0.0207
Total payments (£)	-19.01	84.66
Automatic payments (£)	-2.0348	42.0977
Manual payments (£)	-16.02	67.07
Total payments (% statement balance)	-0.0008	0.0435
Automatic payments (% statement balance)	0.0046	0.0571
Manual payments (% statement balance)	-0.0091	0.0488
Cumulative total payments (£)	-49.33	304.33
Cumulative automatic payments (£)	21.21	249.63
Cumulative manual payments (£)	-89.30	200.32

Notes: Table shows Lee Bounds (Lee, 2009) for the treatment effects on primary outcomes after seven statement cycles (δ_7). To calculate Lee bounds, we adapted public code from Levy (2021) trimming the excess observations and then run regressions. The term δ_7 is one of the coefficients (δ_τ) on interaction terms between treatment indicator and statement cycle indicators in the OLS regression specified in Equation 1. Regressions also include statement cycle fixed effects, year-month fixed effects, and the following controls: Gender, Age, Age squared, Log Estimated Income, Credit Score, Unsecured Debt-to-Income (DTI) Ratio, Any Mortgage Debt, Log Credit Card Credit Limit, Credit Card Purchases Rate, Subprime Credit Card, Any Credit Card Promotional Rate, Any Credit Card Balance Transfer, Credit Card Open Date, Credit Card Statement Day, Any Credit Card Secondary Cardholder. These are all from the time of card origination except for the variables constructed from consumer credit reporting data (Credit Score, DTI Ratio and Any Mortgage Debt), which are from the month preceding card origination. There are 40,708 consumers with 368,162 observations.

Table A10—Heterogeneous treatment effects on I. any Autopay Min enrollment and II. any minimum payment, by pre-trial variables, pooled across all statement cycles

I: Any Autopay Min Enrollment

Variable	Group	Estimate	S.E.	95% C.I.	P Value	N
Credit Score	Low	-0.26	(0.01)	[-0.27, -0.25]	0.0000	183,504
Credit Score	High	-0.22	(0.01)	[-0.23, -0.21]	0.0000	184,658
Income	Low	-0.23	(0.01)	[-0.24, -0.22]	0.0000	183,841
Income	High	-0.24	(0.01)	[-0.25, -0.23]	0.0000	184,321
Unsecured Debt-to-Income (DTI) Ratio	Low	-0.20	(0.00)	[-0.21, -0.19]	0.0000	183,994
Unsecured Debt-to-Income (DTI) Ratio	High	-0.28	(0.01)	[-0.29, -0.27]	0.0000	184,168
Age	Low	-0.22	(0.01)	[-0.23, -0.21]	0.0000	191,854
Age	High	-0.26	(0.01)	[-0.27, -0.25]	0.0000	176,308
Gender	Male	-0.24	(0.01)	[-0.25, -0.23]	0.0000	198,216
Gender	Female	-0.23	(0.01)	[-0.24, -0.22]	0.0000	169,946
Credit Limit	Low	-0.24	(0.01)	[-0.25, -0.23]	0.0000	184,928
Credit Limit	High	-0.24	(0.01)	[-0.25, -0.22]	0.0000	183,234
Number of Credit Cards in Portfolio	Low	-0.22	(0.00)	[-0.23, -0.21]	0.0000	240,492
Number of Credit Cards in Portfolio	High	-0.27	(0.01)	[-0.28, -0.26]	0.0000	127,670
Number of Credit Cards in Portfolio With Debt	Low	-0.21	(0.00)	[-0.22, -0.21]	0.0000	285,382
Number of Credit Cards in Portfolio With Debt	High	-0.31	(0.01)	[-0.33, -0.29]	0.0000	82,780
Credit Card Portfolio Balances Net of Payments	Low	-0.19	(0.00)	[-0.20, -0.18]	0.0000	183,423
Credit Card Portfolio Balances Net of Payments	High	-0.28	(0.01)	[-0.29, -0.27]	0.0000	184,739

II: Any Minimum Payment

Variable	Group	Estimate	S.E.	95% C.I.	P Value	N
Credit Score	Low	-0.09	(0.00)	[-0.10, -0.08]	0.0000	183,504
Credit Score	High	-0.07	(0.00)	[-0.08, -0.06]	0.0000	184,658
Income	Low	-0.08	(0.00)	[-0.09, -0.07]	0.0000	183,841
Income	High	-0.08	(0.00)	[-0.09, -0.07]	0.0000	184,321
Unsecured Debt-to-Income (DTI) Ratio	Low	-0.06	(0.00)	[-0.07, -0.06]	0.0000	183,994
Unsecured Debt-to-Income (DTI) Ratio	High	-0.10	(0.01)	[-0.11, -0.09]	0.0000	184,168
Age	Low	-0.07	(0.00)	[-0.08, -0.06]	0.0000	191,854
Age	High	-0.09	(0.00)	[-0.10, -0.09]	0.0000	176,308
Gender	Male	-0.08	(0.00)	[-0.09, -0.07]	0.0000	198,216
Gender	Female	-0.08	(0.00)	[-0.09, -0.07]	0.0000	169,946
Credit Limit	Low	-0.09	(0.00)	[-0.09, -0.08]	0.0000	184,928
Credit Limit	High	-0.08	(0.01)	[-0.09, -0.07]	0.0000	183,234
Number of Credit Cards in Portfolio	Low	-0.08	(0.00)	[-0.08, -0.07]	0.0000	240,492
Number of Credit Cards in Portfolio	High	-0.09	(0.01)	[-0.10, -0.08]	0.0000	127,670
Number of Credit Cards in Portfolio With Debt	Low	-0.07	(0.00)	[-0.08, -0.06]	0.0000	285,382
Number of Credit Cards in Portfolio With Debt	High	-0.11	(0.01)	[-0.13, -0.10]	0.0000	82,780
Credit Card Portfolio Balances Net of Payments	Low	-0.06	(0.00)	[-0.07, -0.05]	0.0000	183,423
Credit Card Portfolio Balances Net of Payments	High	-0.10	(0.01)	[-0.11, -0.09]	0.0000	184,739

Notes: Table shows heterogeneous treatment effects pooled across statement cycles. Estimated treatment effects from the coefficient (δ) on the treatment indicator in the OLS regression specified in Equation 2. Regressions also include statement cycle fixed effects, year-month fixed effects, and the following controls: Gender, Age, Age squared, Log Estimated Income, Credit Score, Unsecured Debt-to-Income (DTI) Ratio, Any Mortgage Debt, Log Credit Card Credit Limit, Credit Card Purchases Rate, Subprime Credit Card, Any Credit Card Promotional Rate, Any Credit Card Balance Transfer, Credit Card Open Date, Credit Card Statement Day, Any Credit Card Secondary Cardholder. These are all from the time of card origination except for the variables constructed from consumer credit reporting data (Credit Score, DTI Ratio and Any Mortgage Debt), which are from the month preceding card origination. Error bars are 95% confidence intervals. Standard errors are clustered at consumer-level. Each estimate is from a separate regression for subsamples by quartiles of each heterogeneous variable. Heterogeneous variables are calculated from consumer credit reporting data in month preceding credit card opening. N = 40,708 consumers in total across heterogeneous groups.

Table A11—Heterogeneous treatment effects on credit card debt, by pre-trial variables, pooled across all statement cycles

I: Credit card debt measured by statement balance net of payments (% statement balance)

Variable	Quartile	Estimate	S.E.	95% C.I.	P Value	N
Credit Score	Low	-0.0076	(0.0038)	[-0.0150, -0.0002]	0.0433	183,504
Credit Score	High	-0.0036	(0.0039)	[-0.0111, 0.0040]	0.3560	184,658
Income	Low	-0.0040	(0.0040)	[-0.0118, 0.0037]	0.3075	183,841
Income	High	-0.0063	(0.0037)	[-0.0135, 0.0010]	0.0895	184,321
Unsecured Debt-to-Income (DTI) Ratio	Low	-0.0143	(0.0043)	[-0.0228, -0.0059]	0.0009	183,994
Unsecured Debt-to-Income (DTI) Ratio	High	0.0025	(0.0032)	[-0.0038, 0.0088]	0.4372	184,168
Age	Low	-0.0093	(0.0039)	[-0.0169, -0.0017]	0.0161	191,854
Age	High	-0.0013	(0.0037)	[-0.0087, 0.0060]	0.7259	176,308
Gender	Male	-0.0080	(0.0038)	[-0.0156, -0.0005]	0.0363	198,216
Gender	Female	-0.0022	(0.0038)	[-0.0097, 0.0052]	0.5560	169,946
Credit Limit	Low	-0.0112	(0.0043)	[-0.0195, -0.0028]	0.0089	184,928
Credit Limit	High	0.0014	(0.0034)	[-0.0052, 0.0079]	0.6834	183,234
Number of Credit Cards in Portfolio	Low	-0.0095	(0.0036)	[-0.0165, -0.0025]	0.0076	240,492
Number of Credit Cards in Portfolio	High	0.0014	(0.0040)	[-0.0064, 0.0091]	0.7301	127,670
Number of Credit Cards in Portfolio With Debt	Low	-0.0083	(0.0033)	[-0.0147, -0.0019]	0.0109	285,382
Number of Credit Cards in Portfolio With Debt	High	0.0014	(0.0040)	[-0.0065, 0.0094]	0.7221	82,780
Credit Card Portfolio Balances Net of Payments	Low	-0.0126	(0.0045)	[-0.0215, -0.0037]	0.0057	183,423
Credit Card Portfolio Balances Net of Payments	High	0.0008	(0.0028)	[-0.0046, 0.0062]	0.7629	184,739

II: Credit card debt measured by statement balance net of payments (£)

Variable	Group	Estimate	S.E.	95% C.I.	P Value	N
Credit Score	Low	-1.83	(17.52)	[-36.17, 32.51]	0.9168	183,504
Credit Score	High	-1.46	(23.90)	[-48.31, 45.39]	0.9514	184,658
Income	Low	23.07	(16.55)	[-9.38, 55.51]	0.1635	183,841
Income	High	-23.12	(23.95)	[-70.06, 23.82]	0.3344	184,321
Unsecured Debt-to-Income (DTI) Ratio	Low	2.74	(16.81)	[-30.21, 35.69]	0.8705	183,994
Unsecured Debt-to-Income (DTI) Ratio	High	-12.79	(22.91)	[-57.69, 32.12]	0.5767	184,168
Age	Low	-32.19	(17.21)	[-65.93, 1.55]	0.0615	191,854
Age	High	42.05	(24.44)	[-5.85, 89.95]	0.0854	176,308
Gender	Male	0.17	(21.22)	[-41.42, 41.76]	0.9936	198,216
Gender	Female	9.66	(20.66)	[-30.83, 50.15]	0.6401	169,946
Credit Limit	Low	-13.88	(9.63)	[-32.75, 4.99]	0.1493	184,928
Credit Limit	High	25.98	(30.79)	[-34.36, 86.32]	0.3987	183,234
Number of Credit Cards in Portfolio	Low	2.83	(15.55)	[-27.66, 33.31]	0.8558	240,492
Number of Credit Cards in Portfolio	High	5.29	(29.59)	[-52.71, 63.29]	0.8581	127,670
Number of Credit Cards in Portfolio With Debt	Low	3.53	(14.93)	[-25.74, 32.79]	0.8133	285,382
Number of Credit Cards in Portfolio With Debt	High	-5.32	(36.87)	[-77.59, 66.94]	0.8852	82,780
Credit Card Portfolio Balances Net of Payments	Low	-10.37	(14.77)	[-39.31, 18.58]	0.4827	183,423
Credit Card Portfolio Balances Net of Payments	High	5.25	(23.11)	[-40.05, 50.55]	0.8203	184,739

Notes: Table shows heterogeneous treatment effects pooled across statement cycles. Outcome in Panel I is statement balance net of payments, as a percent of statement balance. Outcome in Panel II is statement balance net of payments, measured in £. Estimated treatment effects from the coefficient (δ) on the treatment indicator in the OLS regression specified in Equation 2. Regressions also include statement cycle fixed effects, year-month fixed effects, and the following controls: Gender, Age, Age squared, Log Estimated Income, Credit Score, Unsecured Debt-to-Income (DTI) Ratio, Any Mortgage Debt, Log Credit Card Credit Limit, Credit Card Purchases Rate, Subprime Credit Card, Any Credit Card Promotional Rate, Any Credit Card Balance Transfer, Credit Card Open Date, Credit Card Statement Day, Any Credit Card Secondary Cardholder. These are all from the time of card origination except for the variables constructed from consumer credit reporting data (Credit Score, DTI Ratio and Any Mortgage Debt), which are from the month preceding card origination. Error bars are 95% confidence intervals. Standard errors are clustered at consumer-level. Each estimate is from a separate regression for subsamples by quartiles of each heterogeneous variable. Heterogeneous variables are calculated from consumer credit reporting data in month preceding credit card opening. $N = 40{,}708$ consumers in total across heterogeneous groups.

B. Results for Second Lender

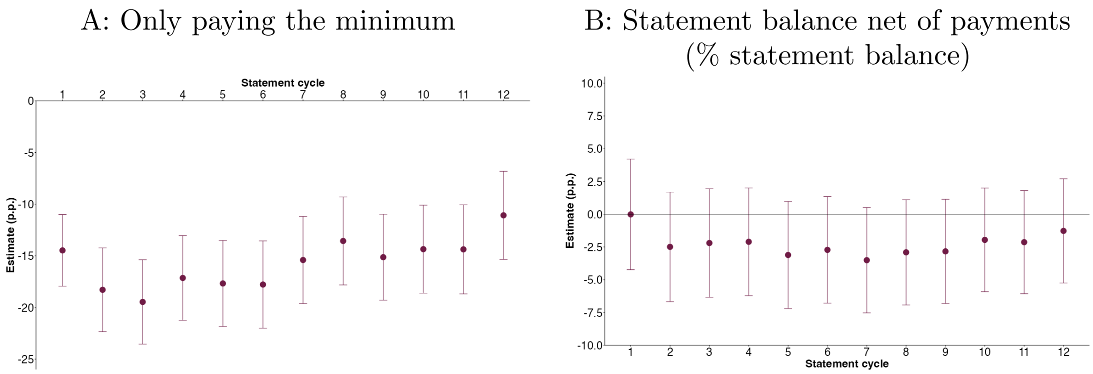

FIGURE B1. SECOND LENDER - AVERAGE TREATMENT EFFECTS ON MAKING ONLY A MINIMUM PAYMENT (PANEL A) AND CREDIT CARD DEBT (PANEL B), ACROSS 1-12 STATEMENT CYCLES

Notes: Outcome in Panel A is primary outcome measure for an indicator of making exactly only the minimum payment, and in Panel B is credit card debt is measured by primary outcome measure: statement balance net of payments (% statement balance). Treatment effects from the coefficients (δ_τ) on interaction terms between treatment indicator and statement cycle indicators in the OLS regression specified in Equation 1. Regressions also include statement cycle fixed effects, year-month fixed effects, and the following controls: Gender, Age, Age squared, Log Estimated Income, Credit Score, Unsecured Debt-to-Income (DTI) Ratio, Any Mortgage Debt, Log Credit Card Credit Limit, Credit Card Purchases Rate, Subprime Credit Card, Any Credit Card Promotional Rate, Any Credit Card Balance Transfer, Credit Card Open Date, Credit Card Statement Day, Any Credit Card Secondary Cardholder. These are all from the time of card origination except for the variables constructed from consumer credit reporting data (Credit Score, DTI Ratio and Any Mortgage Debt), which are from the month preceding card origination. Error bars are 95% confidence intervals. Standard errors are clustered at consumer-level. There are 1,531 credit cards with 19,578 observations.

TABLE B1—SECOND LENDER: BALANCE COMPARISON

Outcome	Control	Treatment	Difference (p.p.)	95% C.I.
Age (years)	37.05	36.48	-0.57	[-1.78, 0.63]
Female (% cards)	0.4774	0.5264	0.0490	[-0.0016, 0.0995]
Any estimated income	0.9248	0.9395	0.0148	[-0.0107, 0.0402]
Estimated income (£)	2073.02	1890.86	-182.16	[-349.54, -14.78]
Credit limit (£)	608.96	587.39	-21.57	[-82.07, 38.93]
Any credit score	0.9863	0.9897	0.0034	[-0.0076, 0.0144]
Credit score (0-100)	0.5369	0.5406	0.0036	[-0.0057, 0.0129]
Purchases rate (%)	22.97	23.46	0.49	[-0.69, 1.67]
Any balance transfer offered	0.1724	0.1699	-0.0025	[-0.0406, 0.0356]
Number of credit cards	2.04	2.00	-0.04	[-0.18, 0.11]
Number of credit cards with debt	0.64	0.63	-0.01	[-0.10, 0.09]
Credit card portfolio statement balances (£)	934.21	872.64	-61.56	[-269.93, 146.80]
Credit card portfolio balances net of payments (£)	855.74	803.06	-52.68	[-249.61, 144.25]

Notes: N (control) = 740 and N (treatment) = 791 cards.

Table B2—Second Lender: Average treatment effects after seven statement cycles

Outcome	Estimate (s.e.)	95% C.I.	P value	Control mean
Any minimum payment	-0.1541 (0.0215)	[-0.1962, -0.1119]	0.0000	0.3160
Any full payment	0.0223 (0.0219)	[-0.0207, 0.0653]	0.3092	0.2503
Any missed payment	0.0089 (0.0170)	[-0.0244, 0.0421]	0.6011	0.1176
Statement balance net of payments (% statement balance)	-0.0351 (0.0205)	[-0.0753, 0.0051]	0.0874	0.6753
Costs (% statement balance)	-0.0089 (0.0040)	[-0.0168, -0.0010]	0.0276	0.0391
Spending (% statement balance)	0.0122 (0.0185)	[-0.0241, 0.0485]	0.5113	0.2245
Share of credit card portfolio only paying minimum	-0.0814 (0.0136)	[-0.1080, -0.0549]	0.0000	0.2016
Share of credit card portfolio making full payment	0.0089 (0.0187)	[-0.0278, 0.0456]	0.6342	0.3455
Share of credit card portfolio missing payment	0.0120 (0.0124)	[-0.0123, 0.0363]	0.3315	0.0904
Credit card portfolio balances net of payments (% statement balances)	-0.0274 (0.0180)	[-0.0627, 0.0078]	0.1276	0.7281
Any autopay	-0.0512 (0.0214)	[-0.0932, -0.0092]	0.0169	0.7606
Autopay full	0.0308 (0.0163)	[-0.0012, 0.0628]	0.0592	0.1081
Autopay fix	0.3036 (0.0229)	[0.2588, 0.3484]	0.0000	0.1860
Autopay min	-0.3856 (0.0209)	[-0.4266, -0.3447]	0.0000	0.4665

Notes: Table shows the treatment effects on primary outcomes after seven statement cycles (δ_7). This treatment effect is one of the coefficients (δ_τ) on interaction terms between treatment indicator and statement cycle indicators in the OLS regression specified in Equation 1. Regressions also include statement cycle fixed effects, year-month fixed effects, and the following controls: Gender, Age, Age squared, Log Estimated Income, Credit Score, Unsecured Debt-to-Income (DTI) Ratio, Any Mortgage Debt, Log Credit Card Credit Limit, Credit Card Purchases Rate, Subprime Credit Card, Any Credit Card Promotional Rate, Any Credit Card Balance Transfer, Credit Card Open Date, Credit Card Statement Day, Any Credit Card Secondary Cardholder. These are all from the time of card origination except for the variables constructed from consumer credit reporting data (Credit Score, DTI Ratio and Any Mortgage Debt), which are from the month preceding card origination. Error bars are 95% confidence intervals. Standard errors are clustered at consumer-level. There are 1,531 credit cards with 19,578 observations.

C. Tertiary Analysis of Liquid Cash Balances

Bank Account Data Sample Restrictions

We keep bank account data on cardholders who appear to be actively using this bank as their primary bank account for a sustained period of time meeting the following criteria: where we observe a solely-held checking account for six months to June 2017, first observed the account at least 180 days before card opening, and where the 3 month moving average of account credits average at least £250 and account debits at least £100 per month during this time (our approach is similar to other research using similar datasets e.g., Ganong and Noel, 2019). For these cardholders we include their liquid cash savings from any other checking accounts held as well as non-checking cash savings accounts with instant access. The choice of threshold used produces similar sample sizes: requiring average account credits and debits are both £500 results in 3,552 cardholders compared to a threshold of £100 that results in 3,831 cardholders. These cardholders are more likely to be younger, with higher incomes and credit scores, fewer credit cards and lower credit card debts as shown in an earlier version of this paper (Guttman-Kenney, 2024).

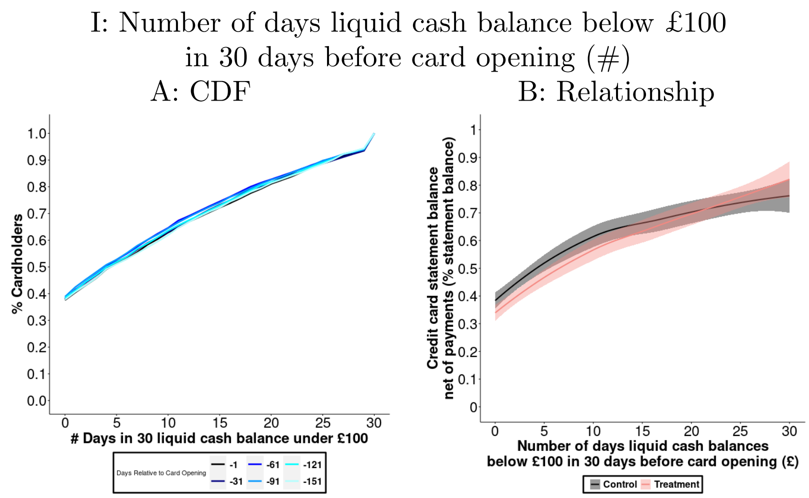

Figure C1. CDF of low liquid cash balances measured before card opening (Panel A) and its non-parametric relationship with credit card debt (statement balance net of payments as a % of statement balance) at statement cycle seven, by treatment group (Panel B)

Notes: N = 3,753 consumers. Panels A is CDF. Panel B is loess. Low Liquid Cash Balance Days shows the number of days when a consumer's liquid cash balances is under £100 in the 30 days before card opening. This is also a dynamic measure where £100 is an arbitrary choice to reflect that not all transactions can be paid with credit cards and therefore consumers may need a positive amount of cash. Different colors in Figure V Panel A show the overall distribution of this variable is stable if calculated at different points-in-time 30 to 150 days before card opening, however, as with our first measure, it may vary for individual consumers over time.

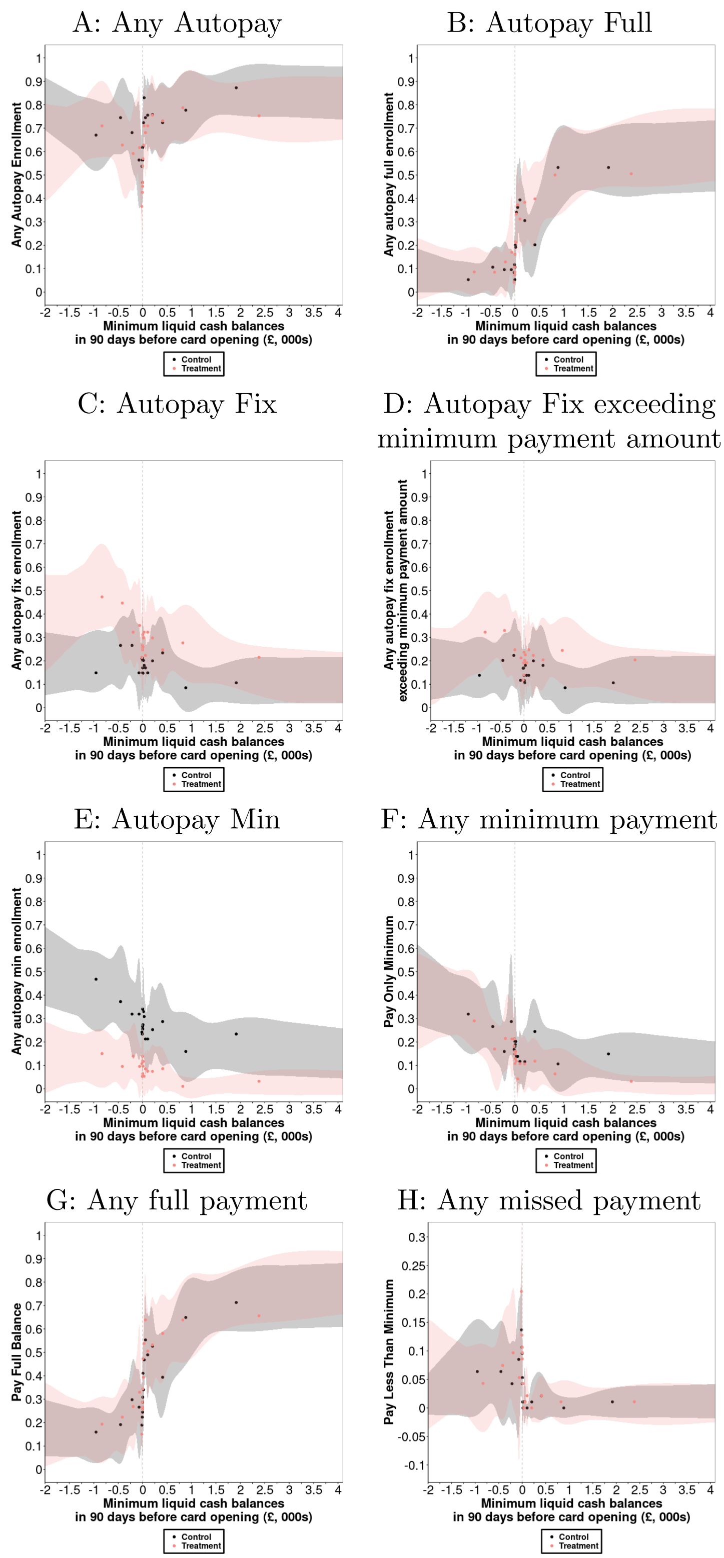

Figure C2. Binned scatterplots of relationship between minimum liquid cash balance during 90 days before card opening with credit card outcomes at statement cycle 7, by treatment group

Notes: N = 3,753 consumers. Liquid cash balances are measured before credit card opening. Panels are binned scatterplots by quantiles of the distribution where error bands are 95% confidence intervals (Cattaneo et al., 2024). X-axes are censored to ease presentation given a fat tail to the distribution of these variables.

Table C1—Summary statistics on I. liquid cash balances by date preceding credit card opening, and II. minimum liquid cash balances over windows preceding credit card opening

I: Liquid cash balances by date preceding credit card opening

Date	Mean	S.D.	P10	P25	P50	P75	P90
-1	2109.85	12324.35	-84.58	48.07	368.65	1,310.91	4,054.58
-31	2142.00	14616.85	-95.17	56.37	364.06	1,297.43	3,757.13
-61	2048.65	9222.26	-61.84	66.93	432.80	1,394.05	4,094.95
-91	2342.60	22005.76	-38.10	66.26	433.57	1,397.41	3,986.56
-121	2164.82	14861.37	-59.16	55.72	396.25	1,401.18	3,949.21
-151	1800.46	7761.59	-75.71	57.62	386.68	1,342.17	3,508.93

II: Minimum liquid cash balances over windows preceding credit card opening

Window	Mean	S.D.	P10	P25	P50	P75	P90
-1 to -31	962.86	5771.79	-487.79	-6.41	24.67	336.62	1,960.99
-1 to -61	780.91	5421.16	-552.73	-14.93	9.50	207.14	1,537.36
-1 to -91	671.38	5107.10	-597.80	-23.85	4.76	142.39	1,296.70
-1 to -121	583.06	4906.39	-629.34	-39.28	2.39	107.63	1,080.03
-1 to -151	485.62	4414.11	-687.15	-51.36	1.08	81.96	909.11

Notes: N = 3,753 consumers. Liquid cash balance is sum of end of day current/checking account and cash saving account balances. Minimum liquid cash balance is minimum value of liquid cash (sum of end of day current/checking account and cash saving account balances) reached by a consumer over 30 to 150 day windows.

Table C2—Correlations with liquid cash balances

Outcome	Liquid Cash Balance	Minimum Liquid Cash Balance	Low Liquid Cash Balance Days
Liquid Cash Balance (£)	0.4720	1	-0.1538
Minimum Liquid Cash Balance (£)	1	0.4720	-0.2083
Low Liquid Cash Balance Days (#)	-0.2083	-0.1538	1
Any minimum payment	-0.0808	-0.0520	0.1545
Any full payment	0.1488	0.1181	-0.3222
Any missed payment	-0.0267	-0.0306	0.1900
Statement balance net of payments (% statement balance)	-0.1554	-0.1213	0.3411
Costs (% statement balance)	-0.0197	-0.0179	0.0622
Spending (% statement balance)	0.0532	0.0274	-0.2091
Share of credit card portfolio only paying minimum	-0.0818	-0.0560	0.1778
Share of credit card portfolio making full payment	0.1406	0.1174	-0.3201
Share of credit card portfolio missing payment	-0.0278	-0.0291	0.1686
Credit card portfolio balances net of payments (% statement balances)	-0.1649	-0.1282	0.3409
Statement balance (£)	-0.1004	-0.0601	0.1633
Statement balance net of payments (£)	-0.1043	-0.0633	0.1764
Cumulative total payments (£)	0.0415	0.0413	-0.0841
Cumulative spending (£)	-0.0427	-0.0144	0.0541
Credit card portfolio statement balances (£)	-0.0770	-0.0378	0.1513
Credit card portfolio balances net of payments (£)	-0.0985	-0.0514	0.1695
Credit score (0-100)	0.0797	0.0803	-0.2347
Estimated income (£)	0.0720	0.1008	-0.0904
Unsecured Debt to Income (DTI) Ratio	-0.0394	-0.0148	0.0603
Age (years)	0.1076	0.1121	-0.0963
Purchases rate (%)	-0.0636	-0.0810	0.1764
Credit limit (£)	0.0576	0.0751	-0.1302

Notes: Table shows correlations between measures of liquid cash balances, outcomes observed after seven completed cycles, and heterogeneous variables observed at card-opening. Liquid cash balances are calculated from bank account data preceding credit card opening. The variable "Minimum Liquid Cash Balance (£)" denotes a heterogeneous cut by the minimum value of liquid cash (sum of end of day current/checking account and cash saving account balances) reached by a consumer in 90 days before card opening. The variable "Minimum Liquid Cash Balance (£)" denotes a heterogeneous cut by the minimum value of liquid cash (sum of end of day current/checking account and cash saving account balances) reached by a consumer in 90 days before card opening. The variable "Liquid Cash Balance

(£)" denotes a heterogeneous cut by the value of liquid cash (sum of end of day current/checking account and cash saving account balances) the day before card opening. The variable "Low Liquid Cash Balance (£)" denotes a heterogeneous cut by the number of days a consumer has under £100 in liquid cash (sum of end of day current/checking account and cash saving account balances) in the 30 days before card opening. N = 3,753 consumers.

TABLE C3—HETEROGENEOUS TREATMENT EFFECTS ON I. ANY AUTOPAY MIN ENROLLMENT AND II. ANY MINIMUM PAYMENT, BY PRE-TRIAL LIQUID CASH BALANCES, POOLED ACROSS ALL STATEMENT CYCLES

I: Any Autopay Min enrollment

Variable	Group	Estimate	S.E.	95% C.I.	P Value	N
Liquid Cash Balance (£)	<£1	-0.3027	(0.0370)	[-0.3752, -0.2302]	0.0000	4,669
Liquid Cash Balance (£)	£1+	-0.1876	(0.0115)	[-0.2102, -0.1650]	0.0000	31,254
Liquid Cash Balance (£)	<£501	-0.2236	(0.0158)	[-0.2545, -0.1927]	0.0000	20,060
Liquid Cash Balance (£)	£501+	-0.1899	(0.0153)	[-0.2199, -0.1600]	0.0000	15,863
Liquid Cash Balance (£)	<£1,001	-0.2166	(0.0140)	[-0.2440, -0.1892]	0.0000	24,970
Liquid Cash Balance (£)	£1,001+	-0.1826	(0.0172)	[-0.2163, -0.1488]	0.0000	10,953
Minimum Liquid Cash Balance (£)	<£1	-0.2574	(0.0193)	[-0.2953, -0.2195]	0.0000	13,434
Minimum Liquid Cash Balance (£)	£1+	-0.1709	(0.0133)	[-0.1969, -0.1448]	0.0000	22,489
Minimum Liquid Cash Balance (£)	<£501	-0.2186	(0.0125)	[-0.2430, -0.1941]	0.0000	30,186
Minimum Liquid Cash Balance (£)	£501+	-0.1159	(0.0209)	[-0.1569, -0.0749]	0.0000	5,737
Minimum Liquid Cash Balance (£)	<£1,001	-0.2133	(0.0121)	[-0.2370, -0.1897]	0.0000	31,841
Minimum Liquid Cash Balance (£)	£1,001+	-0.1236	(0.0211)	[-0.1649, -0.0822]	0.0000	4,082
Low Liquid Cash Balance Days (#)	0 days	-0.1635	(0.0156)	[-0.1942, -0.1329]	0.0000	13,549
Low Liquid Cash Balance Days (#)	1+ days	-0.2264	(0.0148)	[-0.2555, -0.1973]	0.0000	22,374
Low Liquid Cash Balance Days (#)	<16 days	-0.1822	(0.0125)	[-0.2066, -0.1578]	0.0000	25,977
Low Liquid Cash Balance Days (#)	16+ days	-0.2591	(0.0235)	[-0.3053, -0.2130]	0.0000	9,946

II: Any Minimum Payment

Variable	Group	Estimate	S.E.	95% C.I.	P Value	N
Liquid Cash Balance (£)	<£1	-0.0840	(0.0308)	[-0.1443, -0.0236]	0.0066	4,669
Liquid Cash Balance (£)	£1+	-0.0591	(0.0088)	[-0.0764, -0.0419]	0.0000	31,254
Liquid Cash Balance (£)	<£501	-0.0615	(0.0124)	[-0.0858, -0.0371]	0.0000	20,060
Liquid Cash Balance (£)	£501+	-0.0688	(0.0114)	[-0.0912, -0.0464]	0.0000	15,863
Liquid Cash Balance (£)	<£1,001	-0.0549	(0.0109)	[-0.0763, -0.0335]	0.0000	24,970
Liquid Cash Balance (£)	£1,001+	-0.0811	(0.0128)	[-0.1062, -0.0560]	0.0000	10,953
Minimum Liquid Cash Balance (£)	<£1	-0.0794	(0.0160)	[-0.1107, -0.0482]	0.0000	13,434
Minimum Liquid Cash Balance (£)	£1+	-0.0511	(0.0097)	[-0.0702, -0.0320]	0.0000	22,489
Minimum Liquid Cash Balance (£)	<£501	-0.0632	(0.0098)	[-0.0824, -0.0441]	0.0000	30,186
Minimum Liquid Cash Balance (£)	£501+	-0.0522	(0.0162)	[-0.0840, -0.0204]	0.0014	5,737
Minimum Liquid Cash Balance (£)	<£1,001	-0.0638	(0.0094)	[-0.0823, -0.0453]	0.0000	31,841
Minimum Liquid Cash Balance (£)	£1,001+	-0.0493	(0.0173)	[-0.0832, -0.0155]	0.0045	4,082
Low Liquid Cash Balance Days (#)	0 days	-0.0609	(0.0125)	[-0.0854, -0.0364]	0.0000	13,549
Low Liquid Cash Balance Days (#)	1+ days	-0.0620	(0.0115)	[-0.0844, -0.0395]	0.0000	22,374
Low Liquid Cash Balance Days (#)	<16 days	-0.0555	(0.0095)	[-0.0741, -0.0369]	0.0000	25,977
Low Liquid Cash Balance Days (#)	16+ days	-0.0759	(0.0187)	[-0.1126, -0.0392]	0.0001	9,946

Notes: Table shows heterogeneous treatment effects pooled across statement cycles. Outcome in Panel I is any Autopay Min enrollment, and outcome in Panel II is only paying the minimum. Estimated treatment effects from the coefficient (δ) on the treatment indicator in the OLS regression specified in Equation 2. Regressions also include statement cycle fixed effects, year-month fixed effects, and the following controls: Gender, Age, Age squared, Log Estimated Income, Credit Score, Unsecured Debt-to-Income (DTI) Ratio, Any Mortgage Debt, Log Credit Card Credit Limit, Credit Card Purchases Rate, Subprime Credit Card, Any Credit Card Promotional Rate, Any Credit Card Balance Transfer, Credit Card Open Date, Credit Card Statement Day, Any Credit Card Secondary Cardholder. These are all from the time of card origination except for the variables constructed from consumer credit reporting data (Credit Score, DTI Ratio and Any Mortgage Debt), which are from the month preceding card origination. Error bars are 95% confidence intervals. Standard errors are clustered at consumer-level. Each estimate is from a separate regression for subsamples by each heterogeneous variable. Heterogeneous variables are calculated from bank account data preceding credit card opening. The variable "Minimum Liquid Cash Balance (£)" denotes a heterogeneous cut by the minimum value of liquid cash (sum of end of day current/checking account and cash saving account balances) reached by a consumer in 90 days before card opening. The variable "Minimum Liquid Cash Balance (£)" denotes a heterogeneous cut by the minimum value of liquid cash (sum of end of day current/checking account and cash saving account *balances) reached by a consumer in 90 days before card opening. The variable "Liquid Cash Balance (£)" denotes a heterogeneous cut by the value of liquid cash (sum of end of day current/checking account and cash saving account balances) the day before card opening. The variable "Low Liquid Cash Balance (£)" denotes a heterogeneous cut by the number of days a consumer has under £100 in liquid cash (sum of end of day current/checking account and cash saving account balances) in the 30 days before card opening. The column 'Group' denotes the levels of these heterogeneous variables. N = 3,753 consumers in total across heterogeneous groups.*

Table C4—Heterogeneous treatment effects on credit card debt, by pre-trial liquid cash balances, pooled across all statement cycles

I: Credit card debt measured by statement balance net of payments (% statement balance)

Variable	Group	Estimate	S.E.	95% C.I.	P Value	N
Liquid Cash Balance (£)	<£1	-0.0276	(0.0232)	[-0.0730, 0.0178]	0.2337	4,669
Liquid Cash Balance (£)	£1+	-0.0178	(0.0113)	[-0.0399, 0.0043]	0.1152	31,254
Liquid Cash Balance (£)	<£501	-0.0130	(0.0137)	[-0.0398, 0.0138]	0.3412	20,060
Liquid Cash Balance (£)	£501+	-0.0342	(0.0149)	[-0.0635, -0.0049]	0.0221	15,863
Liquid Cash Balance (£)	<£1,001	-0.0115	(0.0124)	[-0.0357, 0.0128]	0.3532	24,970
Liquid Cash Balance (£)	£1,001+	-0.0448	(0.0177)	[-0.0795, -0.0100]	0.0117	10,953
Minimum Liquid Cash Balance (£)	<£1	-0.0181	(0.0158)	[-0.0490, 0.0128]	0.2518	13,434
Minimum Liquid Cash Balance (£)	£1+	-0.0163	(0.0131)	[-0.0421, 0.0094]	0.2142	22,489
Minimum Liquid Cash Balance (£)	<£501	-0.0185	(0.0114)	[-0.0407, 0.0038]	0.1043	30,186
Minimum Liquid Cash Balance (£)	£501+	-0.0302	(0.0221)	[-0.0735, 0.0131]	0.1719	5,737
Minimum Liquid Cash Balance (£)	<£1,001	-0.0210	(0.0111)	[-0.0427, 0.0006]	0.0571	31,841
Minimum Liquid Cash Balance (£)	£1,001+	-0.0163	(0.0252)	[-0.0657, 0.0331]	0.5189	4,082
Low Liquid Cash Balance Days (#)	0 days	-0.0237	(0.0163)	[-0.0556, 0.0082]	0.1457	13,549
Low Liquid Cash Balance Days (#)	1+ days	-0.0132	(0.0128)	[-0.0382, 0.0118]	0.3000	22,374
Low Liquid Cash Balance Days (#)	<16 days	-0.0201	(0.0122)	[-0.0440, 0.0039]	0.1010	25,977
Low Liquid Cash Balance Days (#)	16+ days	-0.0036	(0.0175)	[-0.0379, 0.0307]	0.8368	9,946

II: Credit card debt measured by statement balance net of payments (£)

Variable	Group	Estimate	S.E.	95% C.I.	P Value	N
Liquid Cash Balance (£)	<£1	15.14	(108.17)	[-196.88, 227.16]	0.8888	4,669
Liquid Cash Balance (£)	£1+	30.53	(39.52)	[-46.92, 107.99]	0.4398	31,254
Liquid Cash Balance (£)	<£501	-24.32	(48.70)	[-119.77, 71.13]	0.6176	20,060
Liquid Cash Balance (£)	£501+	54.33	(56.34)	[-56.10, 164.76]	0.3350	15,863
Liquid Cash Balance (£)	<£1,001	24.72	(44.49)	[-62.49, 111.92]	0.5786	24,970
Liquid Cash Balance (£)	£1,001+	-13.32	(66.21)	[-143.09, 116.44]	0.8406	10,953
Minimum Liquid Cash Balance (£)	<£1	-12.45	(70.97)	[-151.54, 126.65]	0.8608	13,434
Minimum Liquid Cash Balance (£)	£1+	44.47	(42.27)	[-38.39, 127.32]	0.2930	22,489
Minimum Liquid Cash Balance (£)	<£501	26.97	(42.28)	[-55.89, 109.83]	0.5236	30,186
Minimum Liquid Cash Balance (£)	£501+	-39.34	(70.57)	[-177.66, 98.98]	0.5774	5,737
Minimum Liquid Cash Balance (£)	<£1,001	21.65	(40.79)	[-58.29, 101.59]	0.5955	31,841
Minimum Liquid Cash Balance (£)	£1,001+	24.16	(81.26)	[-135.11, 183.42]	0.7664	4,082
Low Liquid Cash Balance Days (#)	0 days	43.97	(55.99)	[-65.77, 153.70]	0.4324	13,549
Low Liquid Cash Balance Days (#)	1+ days	0.17	(49.76)	[-97.36, 97.69]	0.9973	22,374
Low Liquid Cash Balance Days (#)	<16 days	37.04	(42.54)	[-46.34, 120.41]	0.3840	25,977
Low Liquid Cash Balance Days (#)	16+ days	-7.10	(73.42)	[-151.00, 136.80]	0.9230	9,946

Notes: Table shows heterogeneous treatment effects pooled across statement cycles. Outcome in Panel I is statement balance net of payments, as a percent of statement balance. Outcome in Panel II is statement balance net of payments, measured in £. Estimated treatment effects from the coefficient (δ) on the treatment indicator in the OLS regression specified in Equation 2. Regressions also include statement cycle fixed effects, year-month fixed effects, and the following controls: Gender, Age, Age squared, Log Estimated Income, Credit Score, Unsecured Debt-to-Income (DTI) Ratio, Any Mortgage Debt, Log Credit Card Credit Limit, Credit Card Purchases Rate, Subprime Credit Card, Any Credit Card Promotional Rate, Any Credit Card Balance Transfer, Credit Card Open Date, Credit Card Statement Day, Any Credit Card Secondary Cardholder. These are all from the time of card origination except for the variables constructed from consumer credit reporting data (Credit Score, DTI Ratio and Any Mortgage Debt), which are from the month preceding card origination. Error bars are 95% confidence intervals. Standard errors are clustered at consumer-level. Each estimate is from a separate regression for subsamples by each heterogeneous variable. Heterogeneous variables are calculated from bank account data preceding credit card opening. The variable "Minimum Liquid Cash Balance (£)" denotes a heterogeneous cut by the minimum value of liquid cash (sum of end of day current/checking account and cash saving account balances) reached by a consumer in 90 days before card opening. The variable "Minimum Liquid Cash Balance (£)" denotes a heterogeneous cut by the minimum value of liquid cash *(sum of end of day current/checking account and cash saving account balances) reached by a consumer in 90 days before card opening. The variable "Liquid Cash Balance (£)" denotes a heterogeneous cut by the value of liquid cash (sum of end of day current/checking account and cash saving account balances) the day before card opening. The variable "Low Liquid Cash Balance (£)" denotes a heterogeneous cut by the number of days a consumer has under £100 in liquid cash (sum of end of day current/checking account and cash saving account balances) in the 30 days before card opening. The column 'Group' denotes the levels of these heterogeneous variables. N = 3,753 consumers in total across heterogeneous groups.*

Table C5—T-tests for difference in treatment effects on credit card debt, high compared to low pre-trial liquid cash, pooled across all statement cycles

I: Credit card debt measured by statement balance net of payments (% statement balance)

Variable	Group	High Liquid Cash (%)	High - Low	95% C.I.	P Value
Liquid Cash Balance (£)	£1+	86.97	0.0098	[0.0085, 0.0111]	0.0000
Liquid Cash Balance (£)	£501+	44.10	-0.0212	[-0.0221, -0.0203]	0.0000
Liquid Cash Balance (£)	£1,001+	30.48	-0.0333	[-0.0343, -0.0323]	0.0000
Minimum Liquid Cash Balance (£)	£1+	62.48	0.0018	[0.0009, 0.0027]	0.0002
Minimum Liquid Cash Balance (£)	£501+	15.93	-0.0117	[-0.0129, -0.0105]	0.0000
Minimum Liquid Cash Balance (£)	£1,001+	11.35	0.0047	[0.0033, 0.0061]	0.0000
Low Liquid Cash Balance Days (#)	0 days	37.60	-0.0105	[-0.0114, -0.0096]	0.0000
Low Liquid Cash Balance Days (#)	<16 days	72.18	-0.0165	[-0.0175, -0.0155]	0.0000

II: Credit card debt measured by statement balance net of payments (£)

Variable	Group	High Liquid Cash (%)	High - Low	95% C.I.	P Value
Liquid Cash Balance (£)	£1+	86.97	15.39	[10.30, 20.49]	0.0000
Liquid Cash Balance (£)	£501+	44.10	78.65	[75.29, 82.01]	0.0000
Liquid Cash Balance (£)	£1,001+	30.48	-38.04	[-41.66, -34.42]	0.0000
Minimum Liquid Cash Balance (£)	£1+	62.48	56.91	[53.29, 60.53]	0.0000
Minimum Liquid Cash Balance (£)	£501+	15.93	-66.31	[-70.50, -62.12]	0.0000
Minimum Liquid Cash Balance (£)	£1,001+	11.35	2.50	[-2.25, 7.26]	0.3022
Low Liquid Cash Balance Days (#)	0 days	37.60	43.80	[40.35, 47.25]	0.0000
Low Liquid Cash Balance Days (#)	<16 days	72.18	44.14	[40.35, 47.92]	0.0000

Notes: Table shows t-tests comparing differences in heterogeneous treatment effects by high liquid cash subsample compared to low liquid cash subsample. Outcome in Panel I is statement balance net of payments, as a percent of statement balance. Outcome in Panel II is statement balance net of payments, measured in £. Data are pooled across statement cycles. Estimated treatment effects from the coefficient (δ) on the treatment indicator in the OLS regression specified in Equation 2. Regressions also include statement cycle fixed effects, year-month fixed effects, and the following controls: Gender, Age, Age squared, Log Estimated Income, Credit Score, Unsecured Debt-to-Income (DTI) Ratio, Any Mortgage Debt, Log Credit Card Credit Limit, Credit Card Purchases Rate, Subprime Credit Card, Any Credit Card Promotional Rate, Any Credit Card Balance Transfer, Credit Card Open Date, Credit Card Statement Day, Any Credit Card Secondary Cardholder. These are all from the time of card origination except for the variables constructed from consumer credit reporting data (Credit Score, DTI Ratio and Any Mortgage Debt), which are from the month preceding card origination. Error bars are 95% confidence intervals. Standard errors are clustered at consumer-level. Each estimate is from a separate regression for subsamples by each heterogeneous variable. Heterogeneous variables are calculated from bank account data preceding credit card opening. The variable "Minimum Liquid Cash Balance (£)" denotes a heterogeneous cut by the minimum value of liquid cash (sum of end of day current/checking account and cash saving account balances) reached by a consumer in 90 days before card opening. The variable "Minimum Liquid Cash Balance (£)" denotes a heterogeneous cut by the minimum value of liquid cash (sum of end of day current/checking account and cash saving account balances) reached by a consumer in 90 days before card opening. The variable "Liquid Cash Balance (£)" denotes a heterogeneous cut by the value of liquid cash (sum of end of day current/checking account and cash saving account balances) the day before card opening. The variable "Low Liquid Cash Balance (£)" denotes a heterogeneous cut by the number of days a consumer has under £100 in liquid cash (sum of end of day current/checking account and cash saving account balances) in the 30 days before card opening. The column 'Group' denotes the high liquid cash level of these heterogeneous variables. The column 'High Liquid Cash (%)' denotes the percentage of consumers in the high liquid cash group. The column 'High - Low' denotes the difference in treatment effects for the high liquid cash group less the low liquid cash group, with column '95% C.I.' and 'P Value' from the t-test of this difference. N = 3,753 consumers in total across heterogeneous groups.